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Rashidi

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(54) **OCCLUSIVE DEVICES WITH THROMBOGENIC INSERTS**

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A61B 17/12 (2006.01)

(52) **U.S. Cl.**
CPC .. ***A61B 17/12113*** (2013.01); ***A61B 17/12177*** (2013.01)

(58) **Field of Classification Search**
CPC A61B 17/12113; A61B 17/12177; A61B 2017/1205; A61B 2090/3966; A61B 17/12031; A61B 17/12172
See application file for complete search history.

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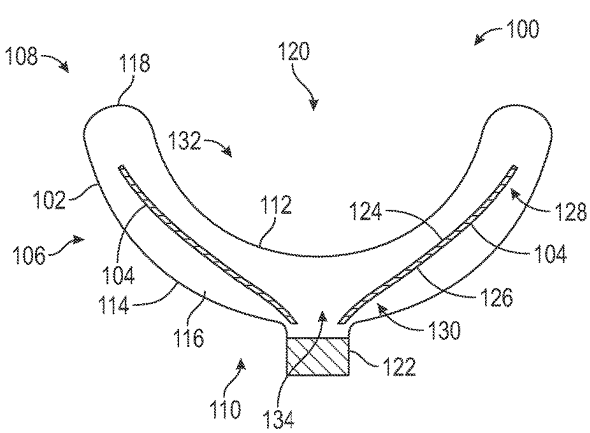
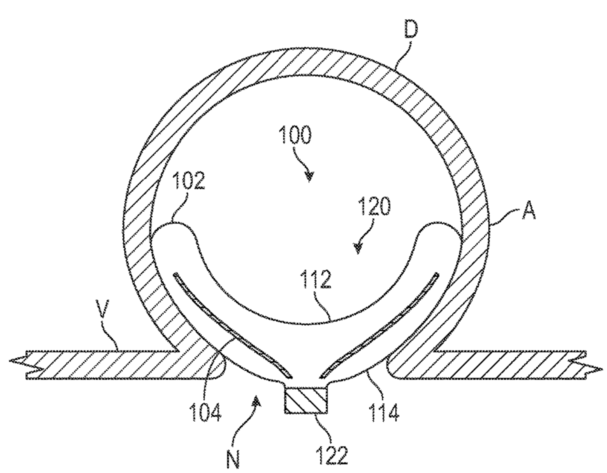
(57) **ABSTRACT**

Devices for treating vascular defects and associated systems and methods are disclosed herein. In some embodiments, an occlusive device for treating an aneurysm includes an expandable mesh configured to span a neck of the aneurysm. The expandable mesh can include an upper wall and a lower wall. The occlusive device can also include an insert positioned within the expandable mesh between the upper and lower walls. The insert can have an inner surface and an outer surface. The inner surface of the insert can face the upper wall of the expandable mesh, and the outer surface of the insert can face the lower wall of the expandable mesh. The insert can be configured to promote thrombosis when the occlusive device is deployed within the aneurysm.

19 Claims, 10 Drawing Sheets





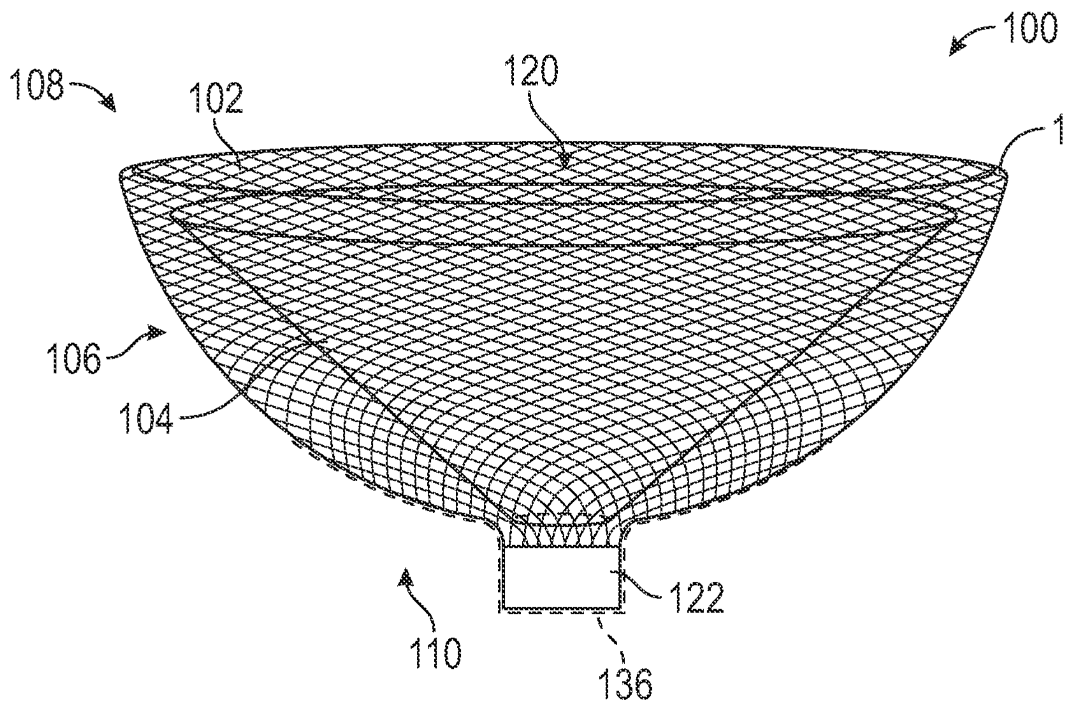

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FIG. 1A

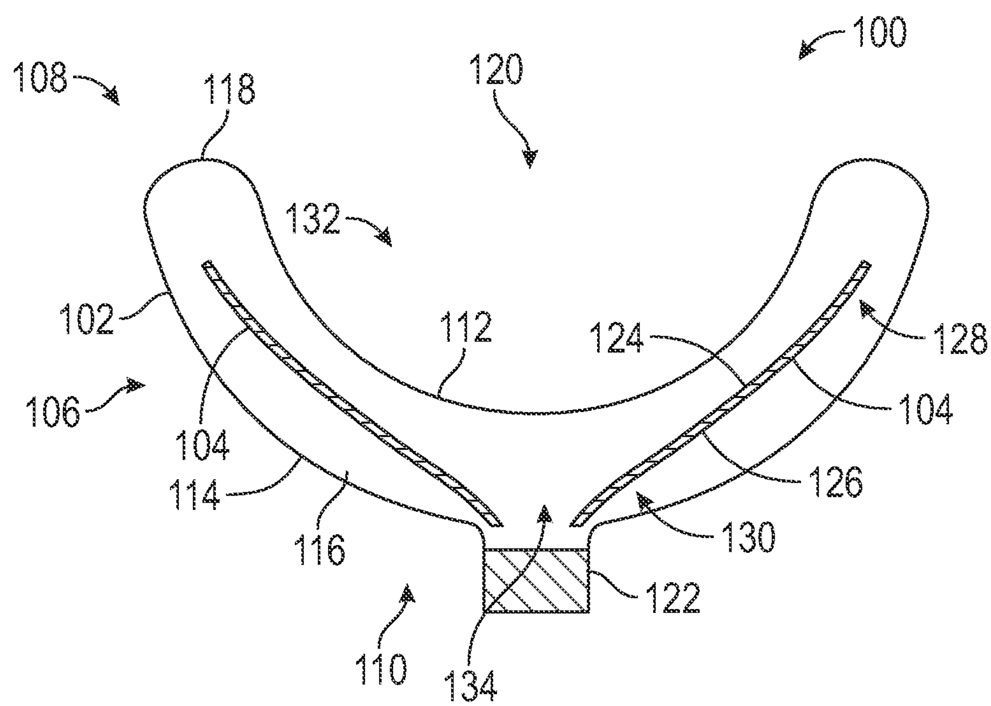
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FIG. 1B

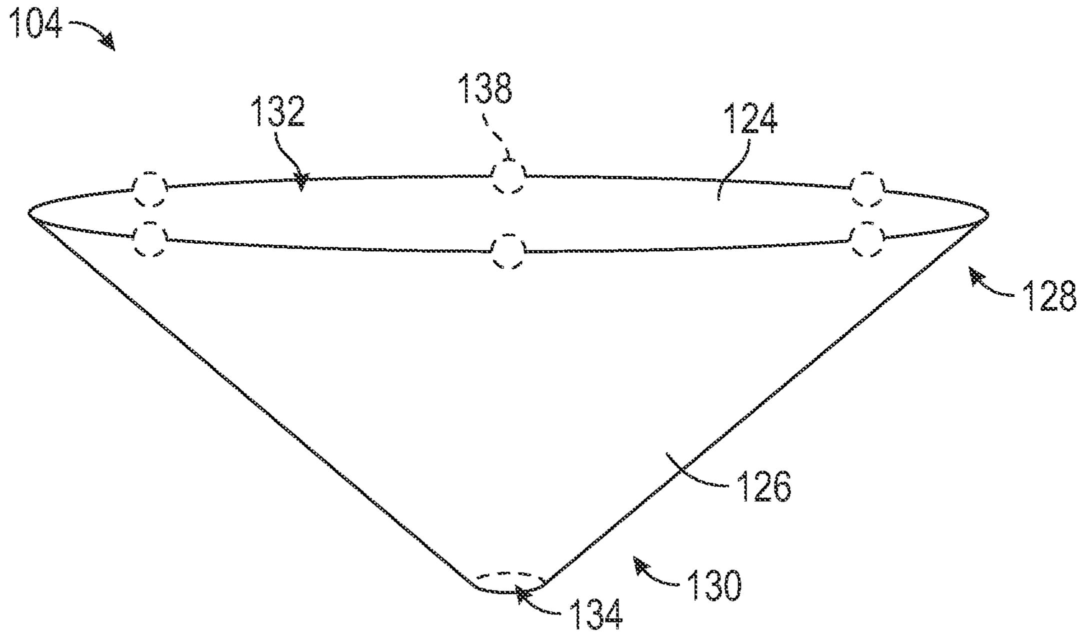

FIG. 1C

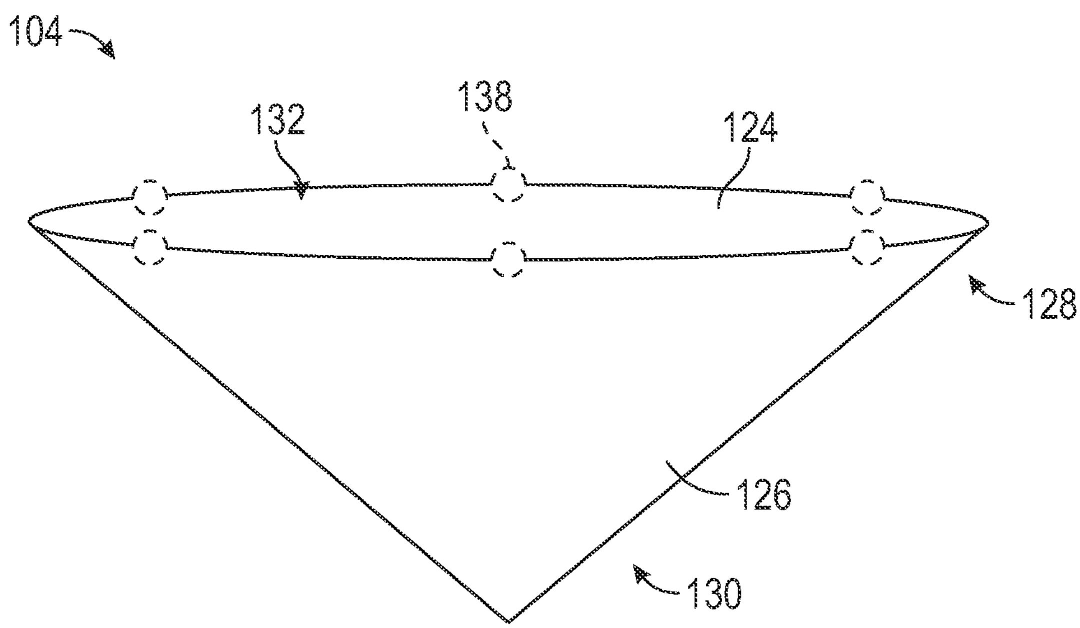

FIG. 1D

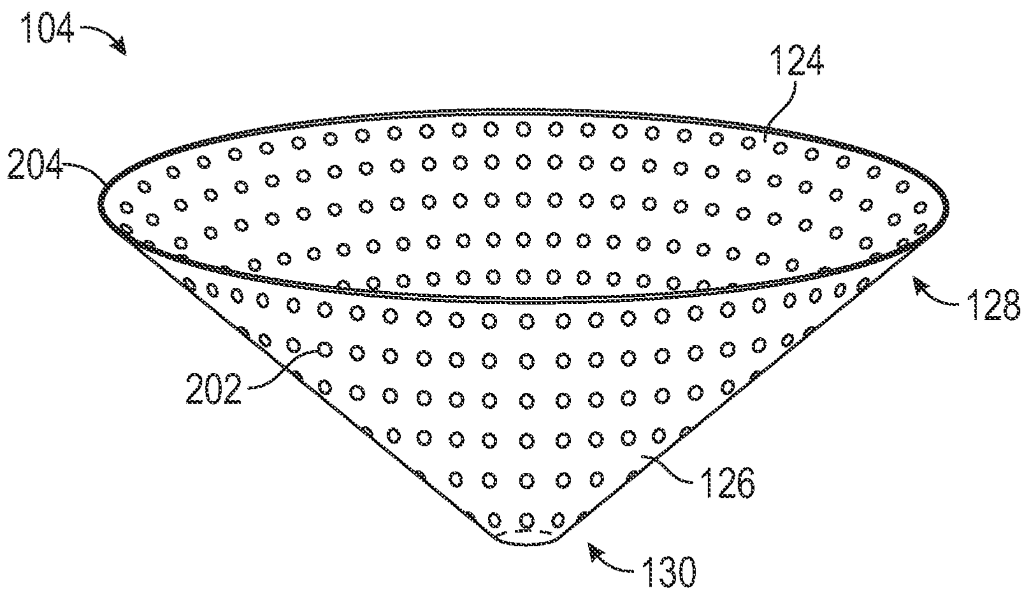

FIG. 2A

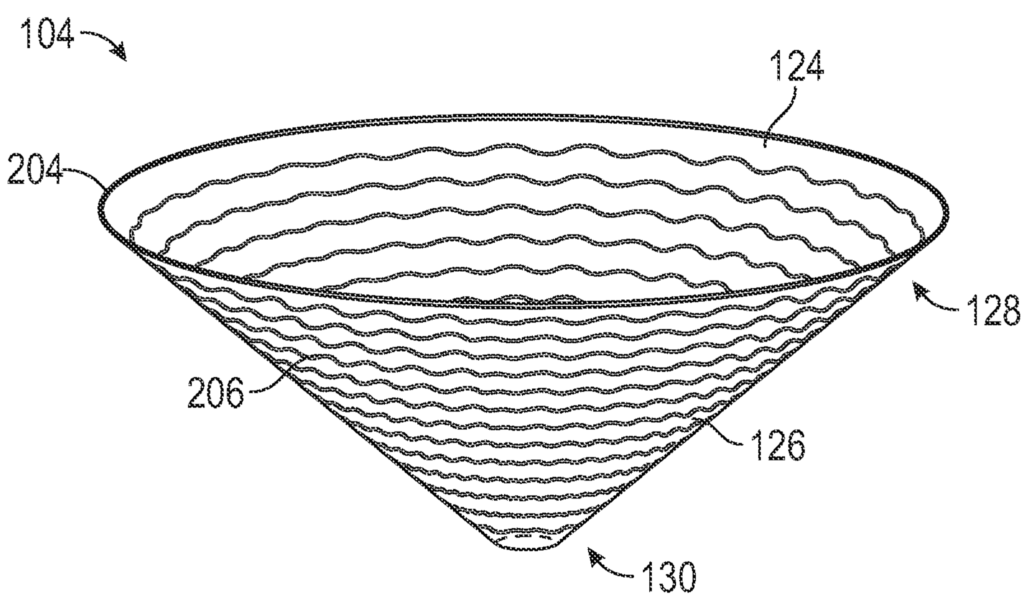

FIG. 2B

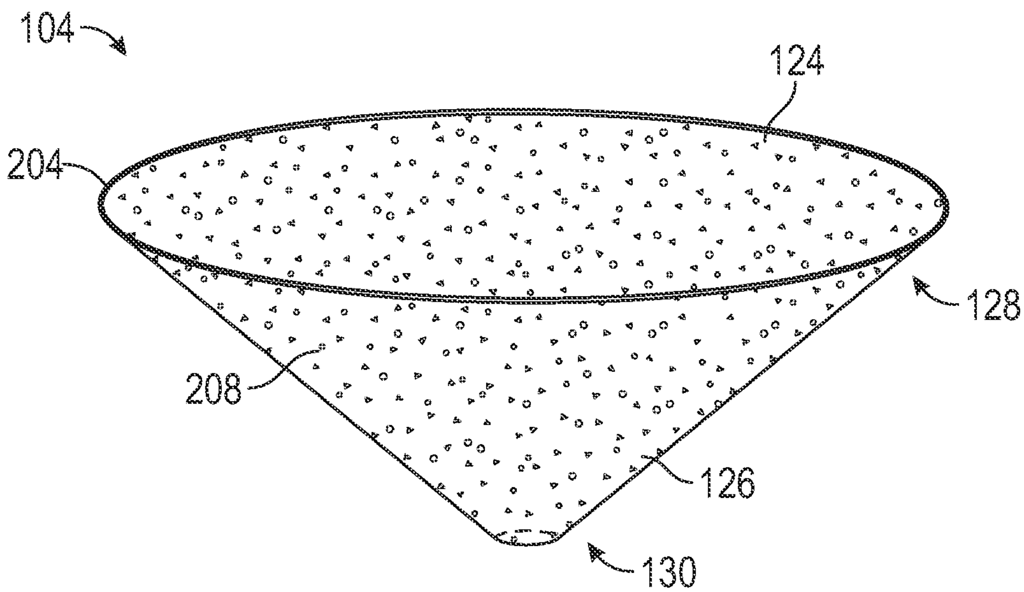

FIG. 2C

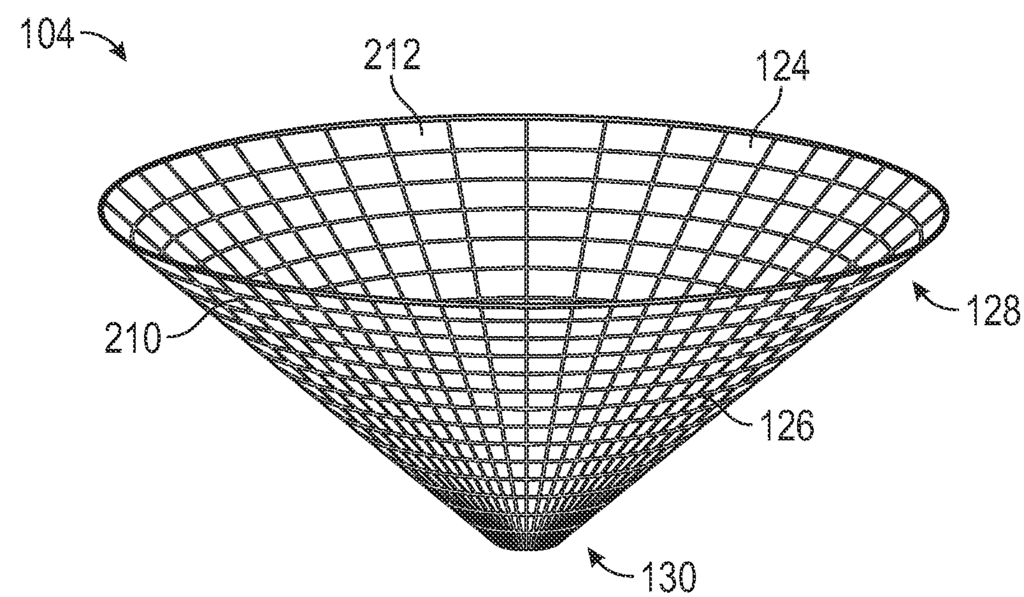

FIG. 2D

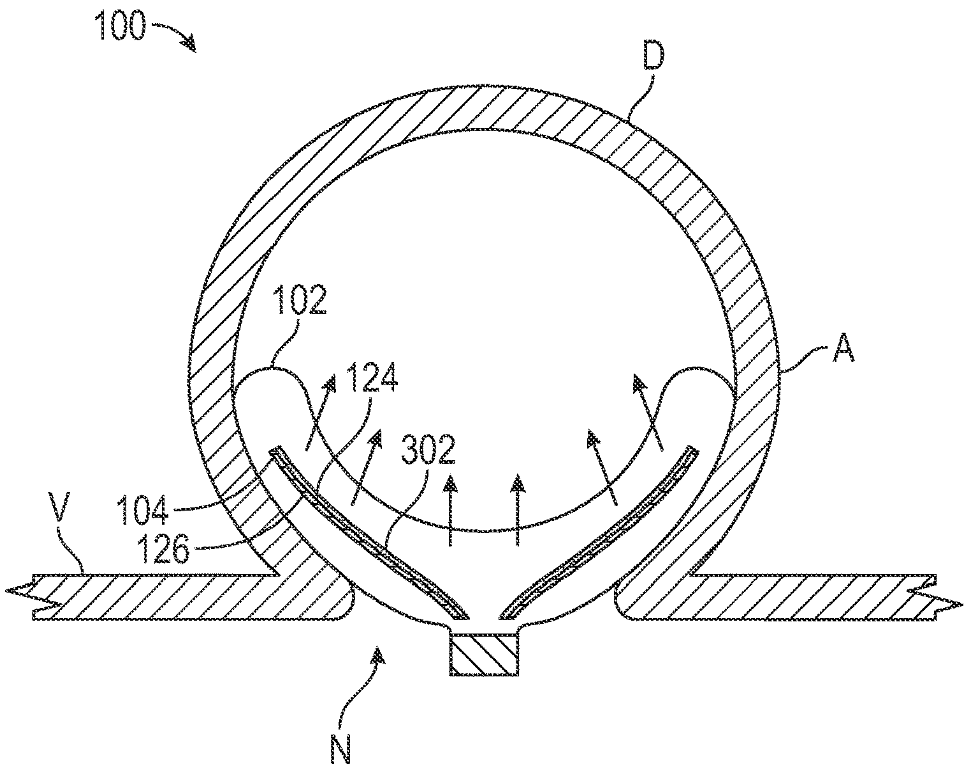

FIG. 3A

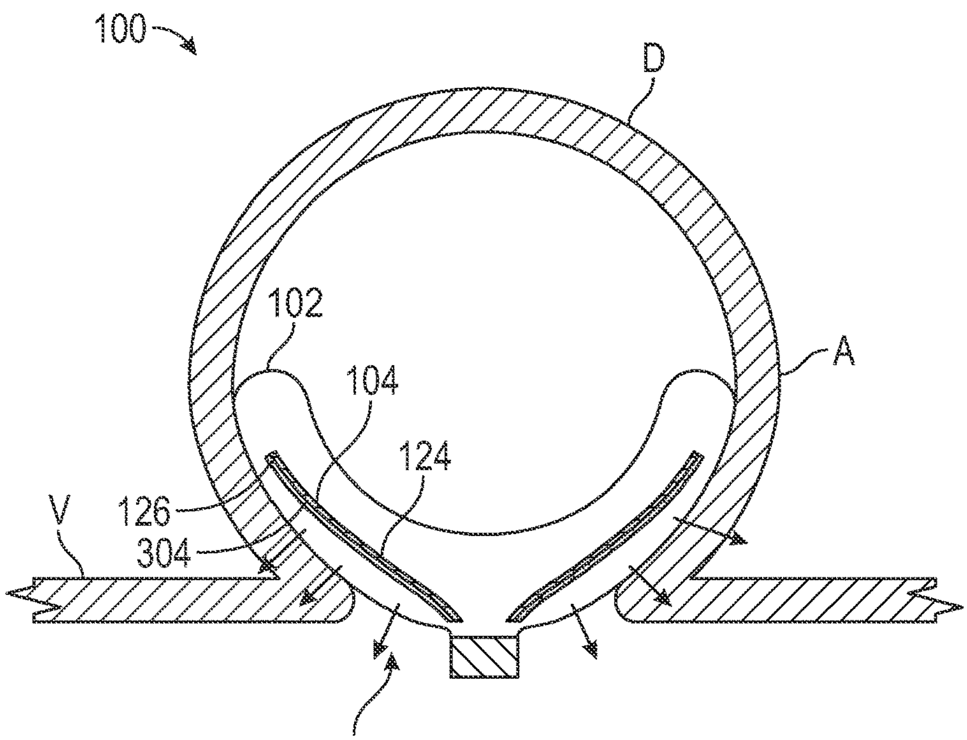

FIG. 3B

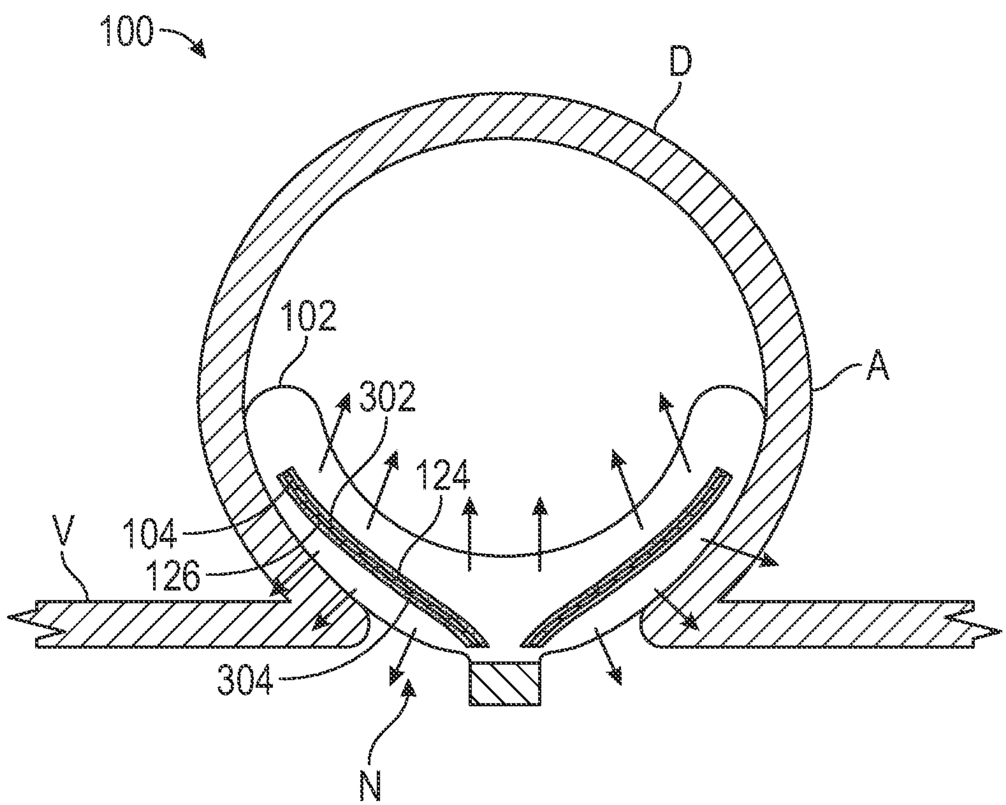

FIG. 3C

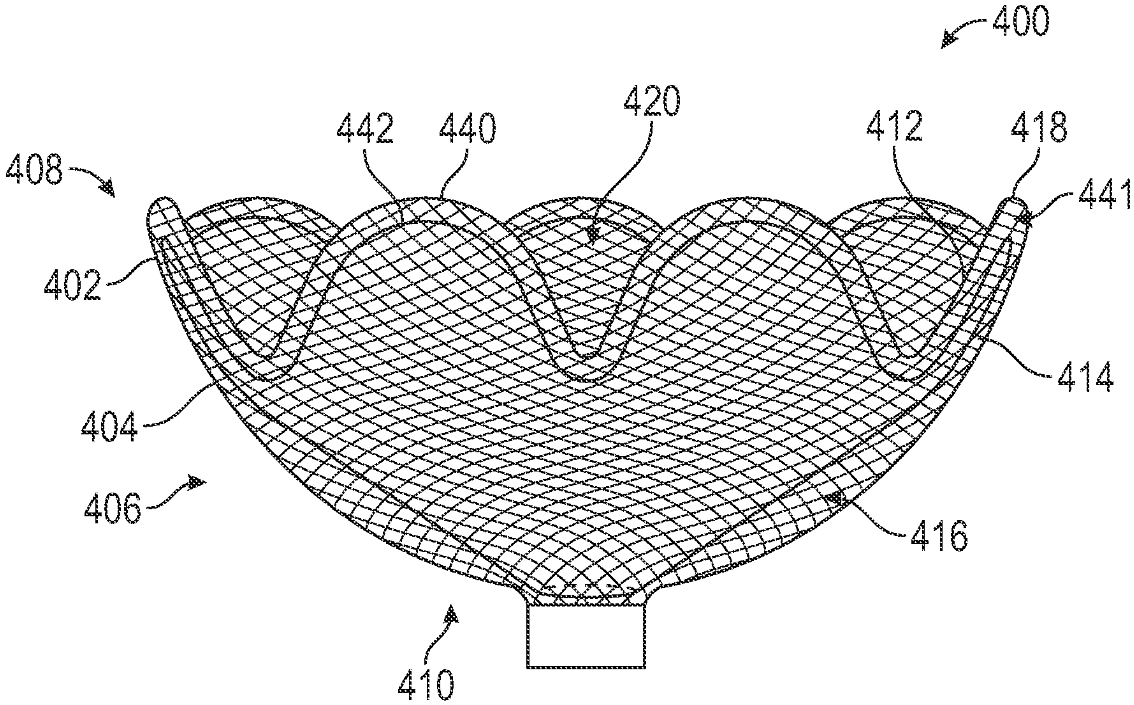
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FIG.4A

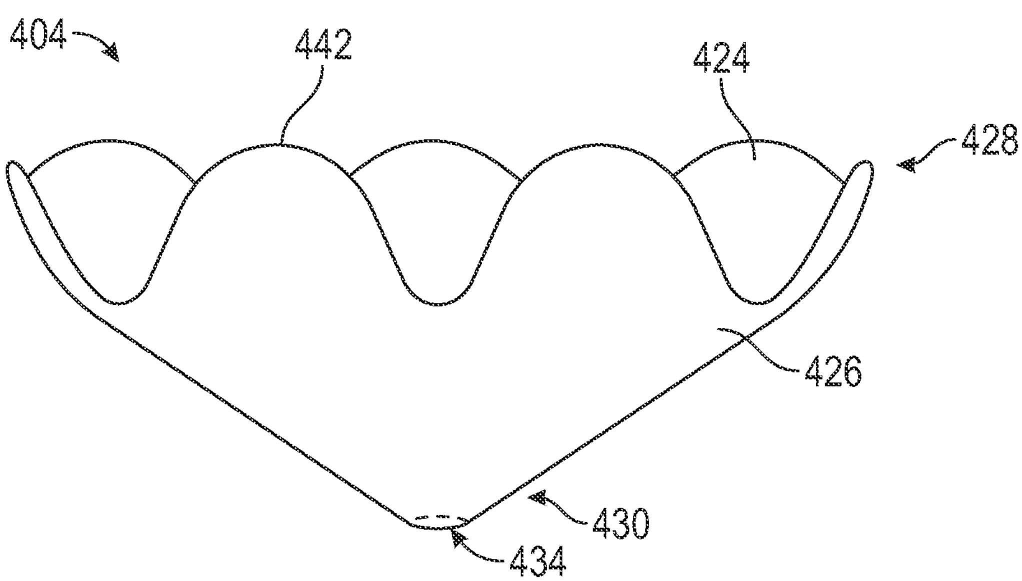
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FIG. 4B

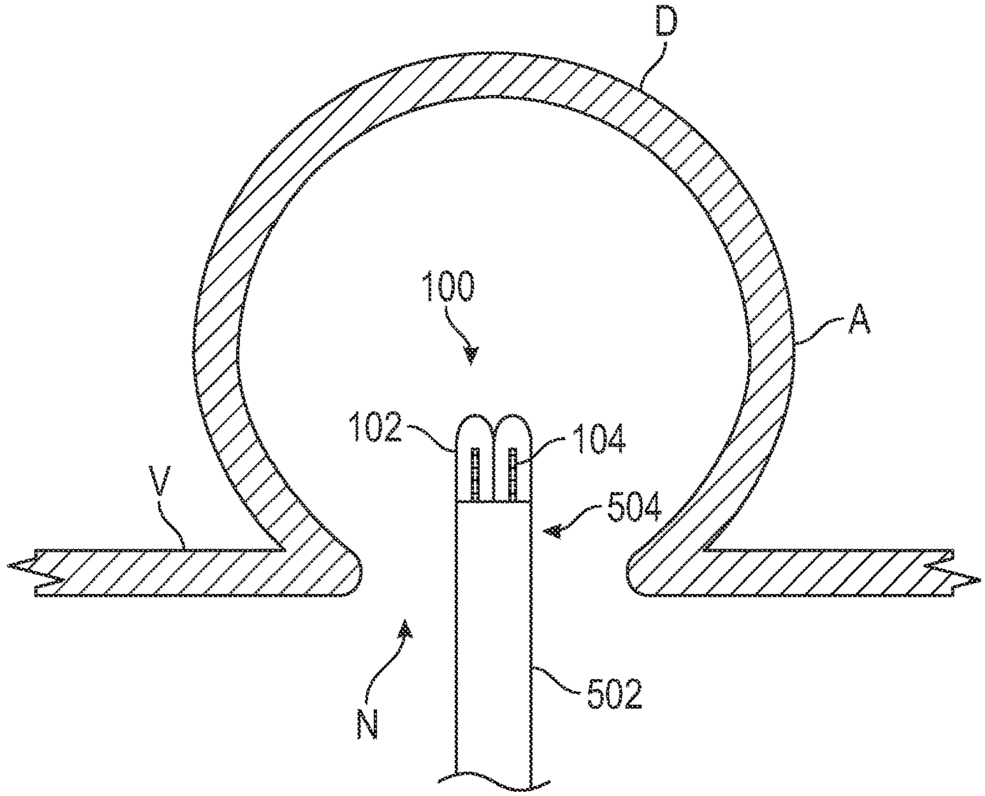

FIG. 5A

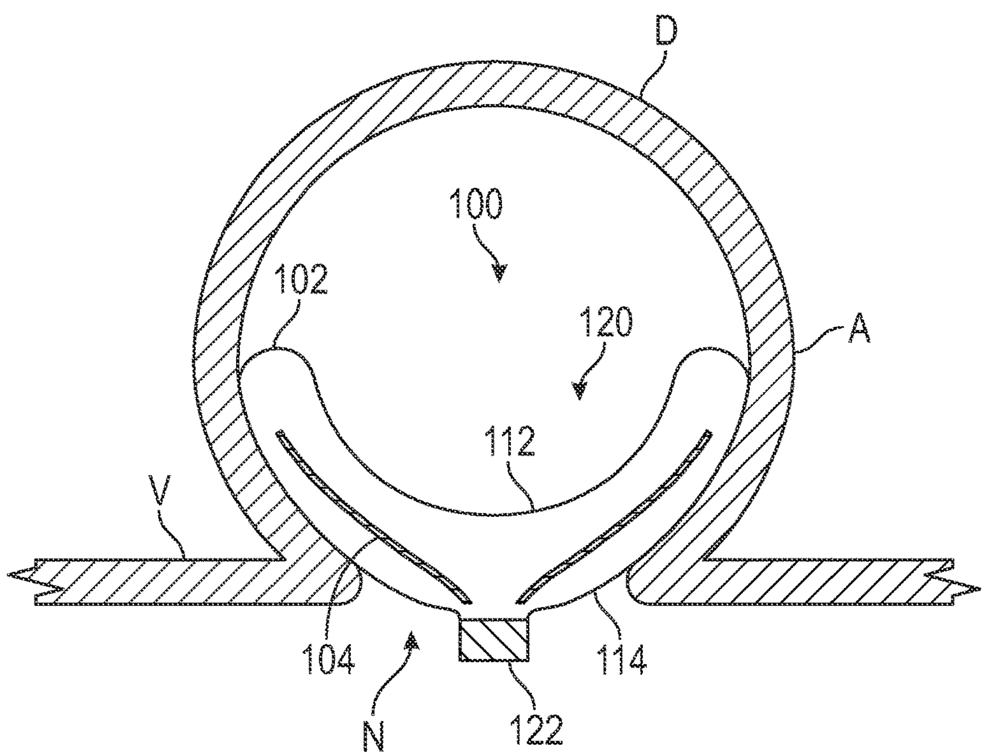

FIG. 5B

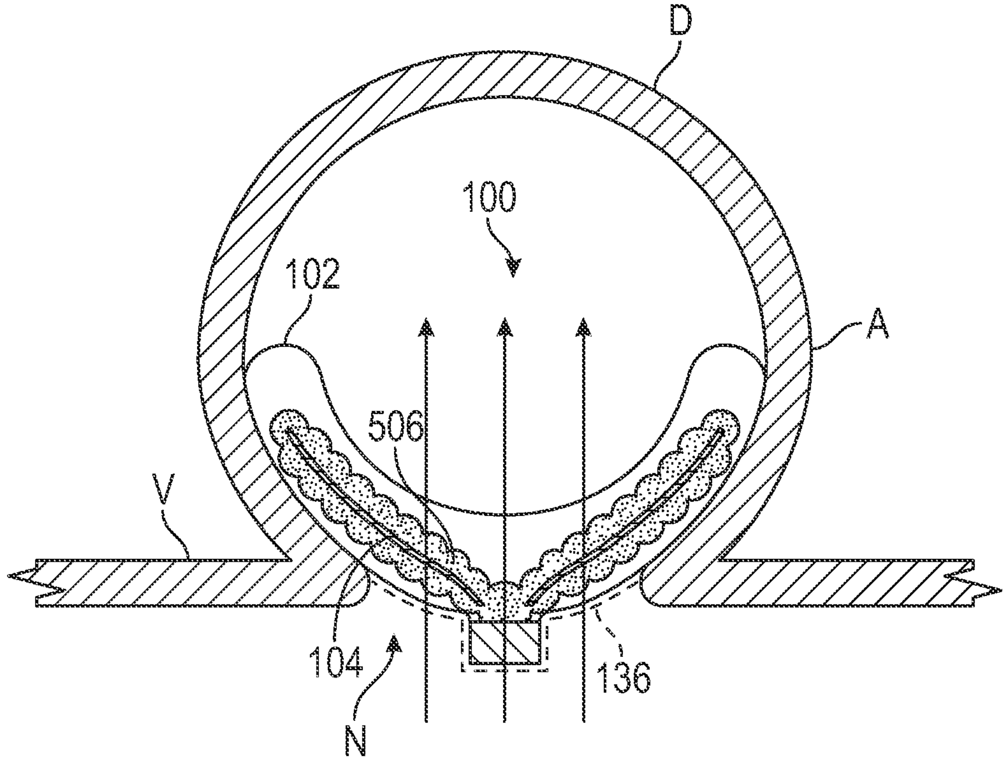

FIG. 5C

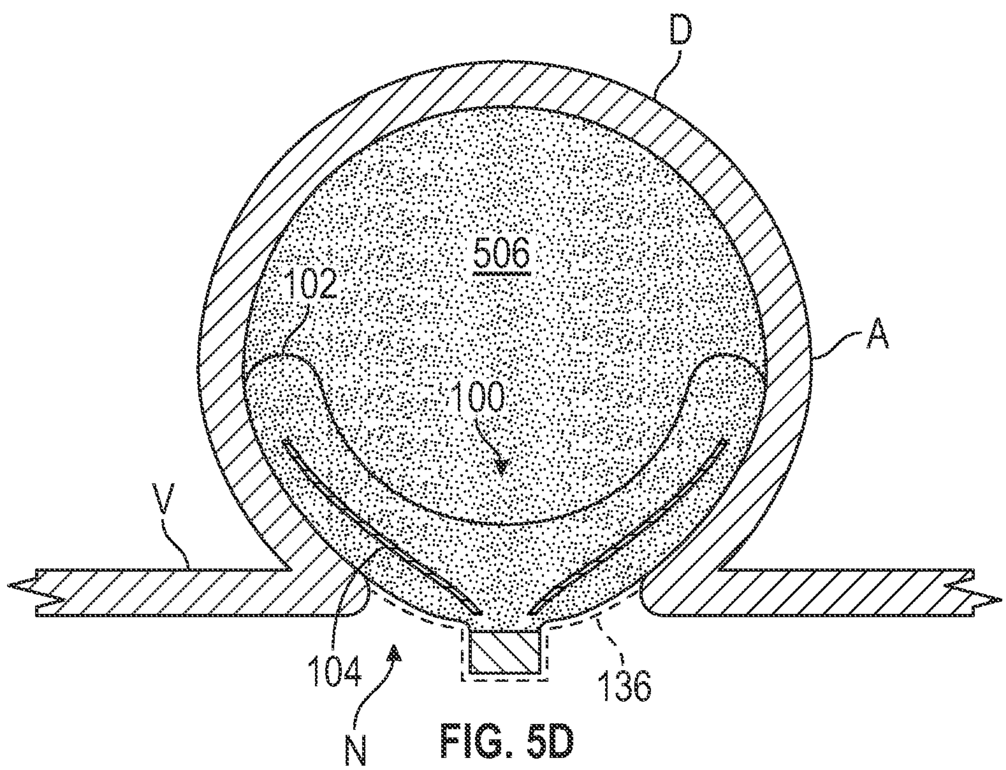

FIG. 5D

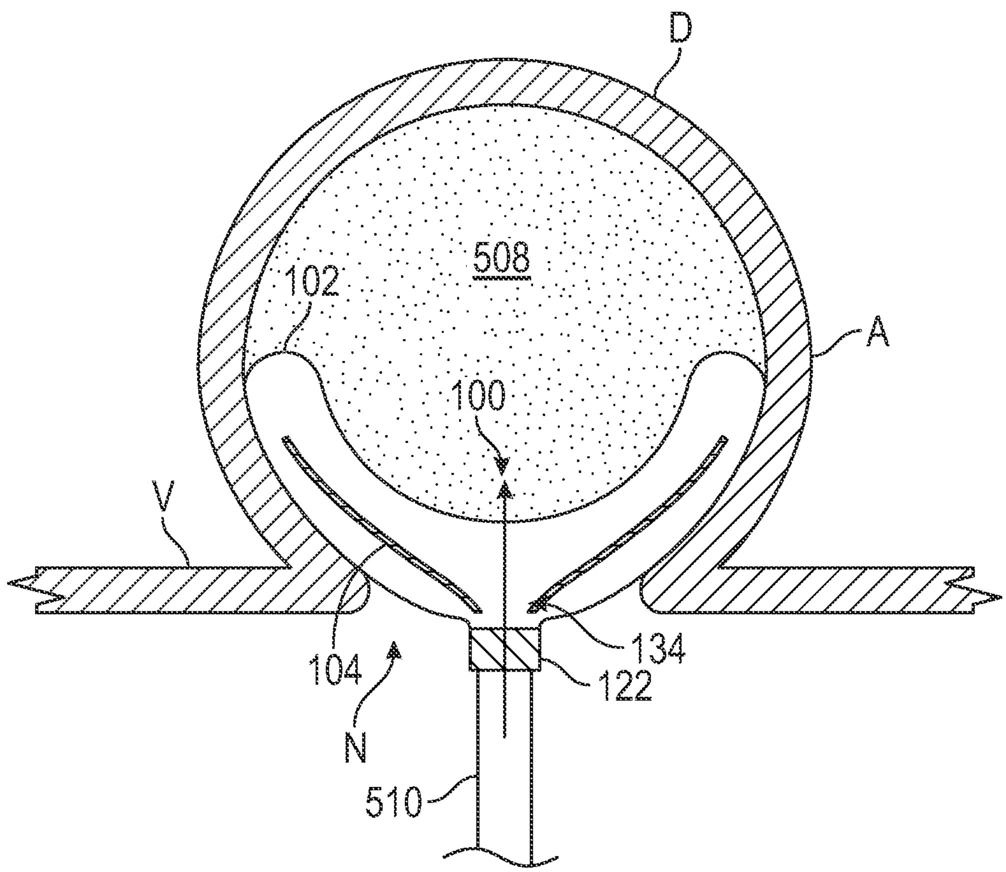

FIG. 5E

OCCLUSIVE DEVICES WITH THROMBOGENIC INSERTS

TECHNICAL FIELD

The present technology generally relates to medical devices, and in particular, to occlusive devices for treating vascular defects.

BACKGROUND

Intracranial saccular aneurysms occur in 1% to 2% of the general population and account for approximately 80% to 85% of non-traumatic subarachnoid hemorrhages. Recent studies show a case fatality rate of 8.3% to 66.7% in patients with subarachnoid hemorrhage. Endovascular treatment of intracranial aneurysms with coil embolization involves packing the aneurysm sac with metal coils to reduce or disrupt the flow of blood into the aneurysm, thereby enabling a local thrombus or clot to form which fills and ultimately closes off the aneurysm. Although coiling has proven to have better outcomes than surgical clipping for both ruptured and unruptured aneurysms, treating complex aneurysms using conventional coiling is challenging. This is especially true for wide-necked aneurysms because coil segments may protrude from the aneurysm sac through the neck of the aneurysm and into the parent vessel, causing serious complications for the patient.

To address this, some treatments include temporarily positioning a balloon within the parent vessel across the neck of the aneurysm to prevent the coils from migrating across the neck during delivery. Alternatively, some treatments include permanently positioning a neck-bridging stent within the parent vessel across the neck of the aneurysm to prevent the coils from migrating across the neck during delivery. While balloon-assisted or stent-assisted coiling for wide-necked aneurysms has shown better occlusion rates and lower recurrence than coiling alone, the recanalization rate of treated large/giant aneurysms can be as high as 18.2%. Moreover, the addition of a balloon or stent and its associated delivery system to the procedure increases the time, cost, and complexity of treatment. Deployment of the stent or balloon during the procedure also greatly increases the risk of an intraprocedural clot forming, and can damage the endothelial lining of the vessel wall. Permanently positioning a stent within the parent vessel increases the chronic risk of clot formation on the stent itself and associated ischemic complications, and thus necessitates the use of dual antiplatelet therapy ("DAPT"). DAPT, in turn, increases the risk and severity of hemorrhagic complications in patients with acutely ruptured aneurysms or other hemorrhagic risks. Thus, neck-bridging stents are not indicated for the treatment of ruptured aneurysms.

The above-noted drawbacks associated with balloon- and stent-assisted coiling techniques influenced the development of intraluminal flow diverting stents, or stent-like structures implanted in the parent vessel across the neck of the aneurysm that redirect blood flow away from the aneurysm, thereby promoting aneurysm thrombosis. Flow diverters have been successfully used for treating wide-necked, giant, fusiform, and blister-like aneurysms. However, because they are positioned in the parent vessel, flow diverters require DAPT to avoid clot formation on the stent itself and ischemic complications. This, in turn, increases the risk and severity of hemorrhagic complications in patients with acutely ruptured aneurysms or other hemorrhagic risks. Thus, flow diverters are not indicated for the treatment of ruptured aneurysms. Flow diverters have also shown limited efficacy in treating bifurcation aneurysms (35-50%).

Endosaccular flow disrupting devices have the potential to provide the intra-aneurysmal flow disruption of coiling with the definitive remodeling at the aneurysm-parent vessel interface achieved by intraluminal flow diverters. Endosaccular devices can be mesh devices configured to be deployed completely within the aneurysm sac, with the interstices of the mesh covering the aneurysm neck and reconstructing the aneurysm-parent vessel interface. The implant disrupts the blood flow entering and exiting the aneurysm sac (resulting in stasis and thrombosis) and supports neoendothelial overgrowth without requiring DAPT (unlike endoluminal flow diverters). Thus, endosaccular devices can be used to treat wide-necked aneurysms and ruptured aneurysms. Moreover, because the device is placed completely within the aneurysm sac, the parent and branch vessels are unimpeded and can be accessed for any further retreatment or subsequent deployment of adjunctive devices during treatment.

Accordingly, there is a need for improved devices, systems, and methods for treating aneurysms and other vascular defects.

SUMMARY

The subject technology is illustrated, for example, according to various aspects described below. These are provided as examples and do not limit the subject technology.

In one aspect of the present technology, an occlusive device for treating an aneurysm is provided. The occlusive device can include an expandable mesh configured to span a neck of the aneurysm, the expandable mesh including an upper wall and a lower wall. An internal cavity can be defined between the upper wall and the lower wall. The occlusive device can also include an insert positioned within the internal cavity of the expandable mesh between the upper wall and the lower wall, the insert including an inner surface and an outer surface. The inner surface of the insert can face the upper wall of the expandable mesh, and the outer surface of the insert can face the lower wall of the expandable mesh. The insert can be configured to promote thrombosis when the occlusive device is deployed within the aneurysm.

In some embodiments, when the occlusive device is deployed within the aneurysm, the expandable mesh forms a bowl structure including a distal portion, a proximal portion, and a cavity extending from the distal portion toward the proximal portion. The upper wall of the expandable mesh can form a concave surface defining the cavity of the bowl structure, and the lower wall of the expandable mesh can form the proximal portion of the bowl structure. When the occlusive device is deployed within the aneurysm, the insert can include a flared shape including a proximal end, and a distal end wider than the proximal end. The distal end of the insert can be positioned near and proximal to the distal portion of the bowl structure, and the proximal end of the insert can be positioned near and distal to the proximal portion of the bowl structure. The distal end of the insert can define a distal aperture and the concave surface of the bowl structure can be received at least partially within the distal aperture. The proximal end of the insert can define a proximal aperture configured to allow material to pass therethrough, or can be closed.

In some embodiments, the distal end of the bowl structure includes a plurality of mesh petals, each mesh petal being defined by a distal end of the upper wall and a distal end of the lower wall of the expandable mesh, and each mesh petal including a distal end sub-cavity between the distal end of the upper wall and the distal end of the lower wall. The distal end sub-cavity can be connected to the internal cavity of the expandable mesh. The distal end of the insert can include a plurality of insert petals. The number of insert petals can be the same as the number of mesh petals. Each insert petal can be radially aligned with and positioned at least partially within the distal end sub-cavity of a corresponding mesh petal.

In some embodiments, the expandable mesh includes a braid.

In some embodiments, the insert includes a polymeric material. The polymeric material can be expanded polytetrafluoroethylene.

In some embodiments, the insert includes a metallic material. The metallic material can be Cobalt Chromium.

In some embodiments, the insert defines a plurality of pores extending between the inner surface and the outer surface of the insert. The plurality of pores can be configured to enhance thrombogenicity of the insert. The plurality of pores can define an average diameter within a range from 10 microns to 300 microns.

In some embodiments, the insert includes texturing on one or more of the inner surface or the outer surface of the insert. The texturing can be configured to enhance thrombogenicity of the insert.

In some embodiments, the insert includes a mesh.

In some embodiments, the insert is configured to release at least one active agent into the aneurysm. The at least one active agent can be configured to promote thrombosis within the aneurysm. The at least one active agent can be incorporated into a first coating on at least one surface of the insert. The first coating can include the at least one active agent combined with a polymeric material. Optionally, the at least one active agent can be incorporated into a bulk material of the insert. The at least one active agent can include a pharmaceutical compound, a protein, a peptide, an antibody, a nucleic acid, a cell, or a particle.

In some embodiments, the insert includes a first portion configured to release a first active agent, and a second portion configured to release a second active agent. The first portion can be the inner surface of the insert, and the second portion can be the outer surface of the insert. The first active agent can be configured to promote thrombosis, and the second active agent can be configured to promote endothelialization.

In some embodiments, the occlusive device further includes a second coating on a portion of the lower wall of the expandable mesh. The second coating can be configured to reduce thrombogenesis on the portion of the lower wall. The second coating can be configured to enhance endothelialization on the portion of the lower wall.

In some embodiments, the occlusive device further includes a hub coupled to the lower wall of the expandable mesh. The hub can include a detachment element configured to releasably couple the expandable mesh to a pusher member.

In another aspect of the present technology, an occlusive device for treating an aneurysm is provided. The occlusive device can include a mesh configured to self-expand into a bowl structure, the mesh including a distal layer and a proximal layer. The distal layer can include a concave surface defining a cavity of the bowl structure. The occlusive device can further include an insert positioned within the mesh between the distal layer and the proximal layer. The insert can include an outer surface and an inner surface. The insert can include at least one thrombogenic feature.

In some embodiments, when the occlusive device is deployed within the aneurysm, the mesh covers a neck of the aneurysm and the cavity faces a dome of the aneurysm.

In some embodiments, the mesh includes a fold region between the distal and proximal layers.

In some embodiments, the outer surface of the insert is oriented toward the proximal layer of the mesh, and the inner surface of the insert is oriented toward the distal layer of the mesh.

In some embodiments, the insert includes a flared shape including a wider distal end and a narrower proximal end. The wider distal end of the insert can be positioned near and proximal to the distal layer of the mesh, and the narrower proximal end of the insert can be positioned near and distal to the proximal layer of the mesh. The wider distal end of the insert can define a distal aperture and the concave surface of the mesh can be received at least partially within the distal aperture. The narrower proximal end of the insert can include a proximal aperture configured to allow material to pass therethrough, or can be closed.

In some embodiments, the mesh includes a braid.

In some embodiments, the at least one thrombogenic feature includes a thrombogenic material. The thrombogenic material can include a polymeric material, a metallic material, or a combination thereof.

In some embodiments, the at least one thrombogenic feature includes a plurality of pores extending between the inner surface and the outer surface of the insert. The plurality of pores can be configured to enhance thrombogenicity of the insert. The plurality of pores can define an average diameter within a range from 0.1 microns to 1000 microns.

In some embodiments, the at least one thrombogenic feature includes texturing on one or more of the inner surface or the outer surface of the insert, wherein the texturing is configured to enhance thrombogenicity of the insert.

In some embodiments, the at least one thrombogenic feature includes a plurality of filaments.

In some embodiments, the at least one thrombogenic feature includes a releasable agent. The releasable agent can be incorporated into a first coating on one or more of the outer surface or the inner surface of the insert. The releasable agent can be incorporated into a bulk material of the insert.

In some embodiments, the occlusive device further includes a second coating on a portion of the proximal layer of the mesh. The second coating can be configured to reduce thrombogenesis on the portion of the proximal layer. The second coating can be configured to enhance endothelialization on the portion of the proximal layer.

In some embodiments, the occlusive device further includes a hub coupled to the proximal layer of the mesh. The hub can include a detachment element configured to releasably couple the mesh to a pusher member.

In a further aspect of the present technology, an occlusive device for treating an aneurysm is provided. The occlusive device can include an expandable mesh configured to span a neck of the aneurysm, the expandable mesh including a distal wall and a proximal wall. The occlusive device can also include an insert positioned between the distal wall and the proximal wall the insert being formed from a continuous material having an inner surface and an outer surface. The insert can include at least one thrombogenic feature.

In some embodiments, the inner surface of the insert faces the distal wall of the expandable mesh, and the outer surface of the insert faces the proximal wall of the expandable mesh.

In some embodiments, when the occlusive device is deployed within the aneurysm, the proximal wall of the expandable mesh covers the neck of the aneurysm. When the occlusive device is deployed within the aneurysm, the expandable mesh can form a bowl structure having a distal portion, a proximal portion, and a cavity extending from the distal portion toward the proximal portion. The distal wall of the expandable mesh can form a concave surface defining the cavity of the bowl structure, and the proximal wall of the expandable mesh can form the proximal portion of the bowl structure. When the occlusive device is deployed within the aneurysm, the insert can have a flared shape with a distal end and a proximal end, the distal end being wider than the proximal end. The distal end of the insert can be positioned near and proximal to the distal portion of the bowl structure, and the proximal end of the insert can be positioned near and distal to the proximal portion of the bowl structure.

In some embodiments, the distal portion of the bowl structure includes a plurality of first flanges. Each first flange can be defined by a distal end of the distal wall and a distal end of the proximal wall of the expandable mesh. Each first flange can include an internal cavity between the distal end of the distal wall and the distal end of the proximal wall. The distal end of the insert can include a plurality of second flanges. The number of second flanges can be the same as the number of first flanges. Each second flange can be radially aligned with and positioned at least partially within the internal cavity of a corresponding first flange

In some embodiments, the expandable mesh includes an annular ridge connecting the distal wall and the proximal wall, the annular ridge defining a distal lip of the expandable mesh. The annular ridge can have a rounded shape configured to avoid tissue damage when the expandable mesh is introduced into the aneurysm.

In some embodiments, the continuous material includes a thrombogenic material, and the at least one thrombogenic feature includes the thrombogenic material. The thrombogenic material can include a polymeric material, a metallic material, or a combination thereof. The continuous material of the insert can include a substrate extending between the inner surface and the outer surface of the insert, and the thrombogenic material can include a thrombogenic agent immobilized on the substrate.

In some embodiments, the at least one thrombogenic feature includes a physical structure formed in the continuous material of the insert. The physical structure can include pores, texturing, protrusions, indentations, or a combination thereof.

In some embodiments, the at least one thrombogenic feature includes a releasable thrombogenic agent. The releasable thrombogenic agent can be incorporated into a coating on one or more of the inner surface or the outer surface of the insert. The releasable thrombogenic agent can be incorporated into the continuous material of the insert.

In some embodiments, the occlusive device further includes an anti-thrombogenic coating on a portion of the proximal wall of the expandable mesh. The occlusive device can also include a hub coupled to the proximal wall of the expandable mesh. The anti-thrombogenic coating can cover at least a portion of the hub.

In another aspect of the present technology, a method of treating an aneurysm is provided. The method can include positioning an occlusive device within the aneurysm. The occlusive device can include a distal mesh layer oriented toward a dome of the aneurysm, a proximal mesh layer covering a neck of the aneurysm, and an insert positioned between the distal mesh layer and the proximal mesh layer, the insert including an inner surface facing the distal mesh layer, and an outer surface facing the proximal mesh layer. The method can also include promoting thrombogenesis within the aneurysm via at least one thrombogenic feature of the insert.

In some embodiments, the method further includes introducing the occlusive device into the aneurysm via an elongate shaft. The occlusive device can be disposed within the elongate shaft in a low-profile configuration. Introducing the occlusive device can include advancing the occlusive device out of the elongate shaft such that the occlusive device self-expands into an expanded configuration. When the occlusive device is in the expanded configuration, the distal mesh layer can include a concave shape forming a cavity facing the dome of the aneurysm. The insert can include a flared shape with a wider distal end oriented toward the dome of the aneurysm and a narrower proximal end oriented toward the neck of the aneurysm.

In some embodiments, the least one thrombogenic feature includes a plurality of pores extending between the inner surface and the outer surface of the insert, the plurality of pores being configured to enhance thrombogenicity of the insert.

In some embodiments, the at least one thrombogenic feature includes texturing on one or more of the inner surface or the outer surface of the insert, the texturing being configured to enhance thrombogenicity of the insert.

In some embodiments, the at least one thrombogenic feature includes a plurality of filaments.

In some embodiments, the at least one thrombogenic feature includes an active agent on or within the insert. The method can further include releasing the active agent from the insert into the aneurysm. Optionally, the method can also include releasing a second active agent from the insert into the aneurysm, the second active agent being different from the active agent. The active agent can be released from a first coating on the inner surface of the insert, and the second active agent can be released from a second coating on the outer surface of the insert.

In some embodiments, the method further includes inhibiting thrombogenesis on a portion of the proximal mesh layer of the occlusive device.

In some embodiments, the method further includes promoting endothelialization on a portion of the proximal mesh layer of the occlusive device.

In some embodiments, the method further includes detaching the occlusive device from a pusher member.

In some embodiments, the method further includes introducing an embolization element into the aneurysm. The embolization element can include a liquid embolic or a coil. The insert can include a lumen, and the embolization element can be introduced into the aneurysm via the lumen. The method can also include retaining the embolization element within the aneurysm via the occlusive device.

Additional features and advantages of the present technology will be set forth in the description below, and in part will be apparent from the description, or may be learned by practice of the present technology. The advantages of the present technology will be realized and attained by the structure particularly pointed out in the written description and claims hereof as well as the appended drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

Many aspects of the present disclosure can be better understood with reference to the following drawings. The components in the drawings are not necessarily to scale. Instead, emphasis is placed on illustrating clearly the principles of the present disclosure.

FIG. **1**A is a perspective view of an occlusive device configured in accordance with embodiments of the present technology.

FIG. **1**B is a side cross-sectional view of the occlusive device of FIG. **1**A.

FIG. **1**C is a perspective view of an insert of the occlusive device of FIG. **1**A.

FIG. **1**D is a perspective of another embodiment of the insert of FIG. **1**C.

FIG. **2**A is a perspective view of an insert including a plurality of pores, in accordance with embodiments of the present technology.

FIG. **2**B is a perspective view of an insert including a plurality of grooves, in accordance with embodiments of the present technology.

FIG. **2**C is a perspective view of an insert including texturing, in accordance with embodiments of the present technology.

FIG. **2**D is a perspective view of an insert formed from a plurality of filaments, in accordance with embodiments of the present technology.

FIG. **3**A is a side cross-sectional view of an occlusive device including an insert with an inner coating, in accordance with embodiments of the present technology.

FIG. **3**B is a side cross-sectional view of an occlusive device including an insert with an outer coating, in accordance with embodiments of the present technology.

FIG. **3**C is a side cross-sectional view of an occlusive device including an insert with an inner coating and an outer coating, in accordance with embodiments of the present technology.

FIG. **4**A is a perspective view of another occlusive device, in accordance with embodiments of the present technology.

FIG. **4**B is a perspective view of an insert of the occlusive device of FIG. **4**A.

FIG. **5**A is a side cross-sectional view of an occlusive device being introduced into an aneurysm, in accordance with embodiments of the present technology.

FIG. **5**B is a side cross-sectional view of the occlusive device of FIG. **5**A after deployment in the aneurysm.

FIG. **5**C is a side cross-sectional view of thrombus formation on an insert of the occlusive device of FIG. **5**B.

FIG. **5**D is a side cross-sectional view of the aneurysm occluded by thrombus and the occlusive device of FIG. **5**B.

FIG. **5**E is a side cross-sectional view of the aneurysm occluded by a liquid embolic and the occlusive device of FIG. **5**B.

DETAILED DESCRIPTION

The present technology relates to devices for treating vascular defects such as aneurysms, and associated systems and methods. In some embodiments, for example, an occlusive device for treating an aneurysm includes an expandable mesh configured to span a neck of the aneurysm. The expandable mesh can be a multilayered structure including an upper wall and a lower wall, with the upper wall oriented toward the aneurysm dome and the lower wall extending across the aneurysm neck. The occlusive device also includes an insert positioned within the expandable mesh between the upper and lower walls. The insert can include at least one feature configured to promote thrombosis when implanted in the aneurysm, referred to herein as a "thrombogenic feature." For example, the thrombogenic feature can include any of the following: (a) physical structures (e.g., pores, indentations, protrusions, texturing, filaments) that increase the blood-contacting surface area of the insert and/or disrupt blood flow near the insert; (b) a thrombogenic material (e.g., expanded polytetrafluoroethylene, Cobalt Chromium); and/or (c) a releasable thrombogenic agent that elutes from the insert into the aneurysm cavity. Accordingly, the occlusive device can promote faster filling and healing of the aneurysm, either as a standalone device or in combination with an embolization element (e.g., a liquid embolic or coil).

Optionally, the occlusive device can be configured to reduce or inhibit thrombosis at locations that pose an embolism risk. For example, the portion of the lower wall of the expandable mesh that spans the neck of the aneurysm can be coated with an anti-thrombogenic material to reduce the likelihood of clot formation. Alternatively or in combination, the insert can be configured to elute an active agent toward the aneurysm neck to inhibit thrombosis and/or promote endothelialization at the neck. This approach can improve the safety of the treatment procedure, as well as enhance the natural healing response.

Embodiments of the present disclosure will be described more fully hereinafter with reference to the accompanying drawings in which like numerals represent like elements throughout the several figures, and in which example embodiments are shown. Embodiments of the claims may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. The examples set forth herein are non-limiting examples and are merely examples among other possible examples.

As used herein, the terms "vertical," "lateral," "upper," and "lower" can refer to relative directions or positions of features of the embodiments disclosed herein in view of the orientation shown in the Figures. For example, "upper" or "uppermost" can refer to a feature positioned closer to the top of a page than another feature. These terms, however, should be construed broadly to include embodiments having other orientations, such as inverted or inclined orientations where top/bottom, over/under, above/below, up/down, and left/right can be interchanged depending on the orientation.

FIGS. **1**A-**1**D illustrate an occlusive device **100** ("device **100**") configured in accordance with embodiments of the present technology. Referring first to FIGS. **1**A and **1**B together, which are perspective and side cross-sectional views of the device **100**, respectively, the device **100** includes a mesh **102** enclosing an insert **104**. The mesh **102** is configured to be deployed within an aneurysm sac to reduce or prevent blood flow from the parent vessel into the aneurysm sac, and/or provide a scaffold for endothelial cell attachment, while the insert **104** is configured to promote thrombosis within the aneurysm sac to fill the interior space. The growth and development of an endothelial layer over the aneurysm neck can wall off the aneurysm from the parent vessel and allow flow dynamics to equilibrate at the defect. Upon endothelialization, the fluid pressure can be evenly distributed along the parent vessel in a manner that inhibits recanalization at the defect post-treatment. Moreover, blood from within the parent vessel no longer has access to the walled-off defect once the endothelialization process is complete and/or the interior of the aneurysm has been occluded by the device **100** and clot material. The device **100** can be used as a standalone device, or can be used to retain an embolization element (e.g., an embolic liquid or coil) within the aneurysm and prevent prolapse into the parent vessel. Accordingly, the device **100** can facilitate healing of the defect and/or prevent recanalization.

The mesh **102** includes an annular body **106** configured to be positioned within the aneurysm sac, a distal portion **108** configured to be oriented toward the aneurysm dome, and a proximal portion **110** configured to cover the aneurysm neck. As best seen in FIG. 1B, the body **106** of the mesh **102** can be a hollow structure including an upper wall **112** (e.g., an upper and/or distal mesh layer), a lower wall **114** (e.g., a lower and/or proximal mesh layer), and an internal cavity **116** defined between the upper wall **112** and lower wall **114**. In some embodiments, the upper wall **112** is curved toward the lower wall **114**, such that the upper wall **112** and lower wall **114** are connected to each other via a fold region **118**. Thus, the fold region **118** can constitute the distal portion **108** of the mesh **102**, and the lower wall **114** can constitute the proximal portion **110** of the mesh **102**. In other embodiments, however, the upper wall **112** can be vertically aligned with the fold region **118**, or can be curved away from the lower wall **114**, such that the upper wall **112** constitutes the distal portion **108** of the mesh **102**.

In some embodiments, the mesh **102** is an expandable element that is configured to transform from a low-profile (e.g., compressed and/or constrained) configuration for delivery to the aneurysm within an elongate shaft (e.g., a delivery catheter), and an expanded configuration (shown in FIGS. **1**A and **1**B) for deployment within the aneurysm. The mesh **102** can be formed of a resilient material and shape set such that, when unconstrained, the mesh **102** self-expands to a predetermined shape that conforms to the interior of the aneurysm. For example, in the illustrated embodiment, the mesh **102** forms a bowl structure when expanded. The upper wall **112** of the mesh **102** can serve as the inner concave surface of the bowl structure and can define a cavity **120** extending from the distal portion **108** toward the proximal portion **110**. The cavity **120** can have a hemispherical, hemi-ellipsoidal, or other suitable shape for receiving and retaining an embolization element within the aneurysm, as described further below. The lower wall **114** can serve as the outer convex surface of the bowl structure and can be configured to apply an outward force against the walls of the aneurysm upon deployment to secure the mesh **102** within the aneurysm. The fold region **118** can form an annular ridge extending circumferentially around the cavity **120** to serve as the distal lip of the bowl structure. The fold region **118** can have a rounded, atraumatic shape to avoid tissue damage when introducing the mesh **102** into the aneurysm.

In other embodiments, however, the mesh **102** can assume a different shape when expanded. Examples of other suitable shapes include, but are not limited to, tubes, cylinders, hemispheres, polyhedrons (e.g., cuboids, tetrahedrons (such as pyramids), octahedrons, prisms), prolate spheroids, oblate spheroids, ovoids, plates (e.g., discs, polygonal plates), non-spherical surfaces of revolution (e.g., toruses, cones, cylinders, or other shapes rotated about a center point or a coplanar axis), or combinations thereof. In such embodiments, the configuration of the upper wall **112**, lower wall **114**, and fold region **118** described herein can be modified as appropriate to form the desired shape. For example, in embodiments where the mesh **102** has a spheroid or ovoid shape, the upper wall **112** can form a convex surface that is curved away from the lower wall **114**, and the fold region **118** can be omitted.

In some embodiments, the mesh **102** is a braid formed of a plurality of braided or woven filaments (e.g., metal wires, polymer wires, or both). For example, the mesh **102** can be formed of 24, 32, 36, 48, 64, 72, 96, 128, or 144 filaments. The filaments can have shape memory and/or superelastic properties, such that the mesh **102** can be heat-set to assume the predetermined shape when released from the constraints of the elongate shaft. In some embodiments, some or all of the filaments (e.g., at least 25%, 50%, 80%, or 100% of the filaments) are made of one or more shape memory and/or superelastic materials (e.g., Nitinol). The filaments can have any suitable size, such as a diameter within a range from 0.0004 inches to 0.0020 inches, or from 0.0009 inches to 0.0012 inches. For example, some or all of the filaments can have a diameter of no more than 0.0004 inches, 0.0005 inches, 0.0006 inches, 0.0007 inches, 0.0008 inches, 0.0009 inches, 0.001 inches, 0.0011 inches, 0.0012 inches, 0.0013 inches, 0.0014 inches, 0.0015 inches, 0.0016 inches, 0.0017 inches, 0.0018 inches, 0.0019 inches, or 0.0020 inches. Some or all of the filaments of the mesh **102** can have the same diameter, or some or all of the filaments can have different cross-sectional diameters. For example, some of the filaments can have a slightly thicker diameter to impart additional strength to the mesh **102**. The thicker filaments can impart greater strength to the mesh **102** without significantly increasing the delivery profile of the device **100**, while the thinner wires can offer some strength while filling out the matrix density of the mesh **102**. In other embodiments, however, the mesh **102** can be a non-braided structure, such as a stent, and the filaments can instead be formed via laser-cutting, etching, or other suitable techniques known to those of skill in the art.

The proximal portion **110** of the mesh **102** can be coupled to a hub **122**. The hub **122** can be a collar, band, ring, etc., that crimps the filaments of the mesh **102** together. The hub **122** can include or be connected to a detachment element (not shown) that releasably couples the device **100** to a pusher member. The pusher member can be an elongate rod, shaft, wire, etc., that is configured to push the device **100** through a distal end of a delivery catheter to deploy the device **100** within the aneurysm, as described further below. Optionally, the pusher member can also be used to pull the device **100** partially or fully back into the delivery catheter, e.g., for repositioning purposes. The detachment element can utilize any suitable detachment technique known to those of skill in the art, such as electrolytic detachment, mechanical detachment, thermal detachment, electromagnetic detachment, or combinations thereof. An example of a detachment element suitable for use with the present technology is the Axium™ or Axium™ Prime Detachable Coil System (Medtronic).

The insert **104** is positioned within the mesh **102** in the internal cavity **116** between the upper wall **112** and lower wall **114**. As best seen in FIG. **1**C, which shows a perspective view of the insert **104**, the insert **104** includes a flattened element (e.g., a sheet, film, strip, mesh, membrane, etc.) that is formed into a three-dimensional structure having an inner surface **124**, an outer surface **126**, a distal end **128**, and a proximal end **130**. The insert **104** can conform to the shape of the internal cavity **116** of the mesh **102**, with the inner surface **124** of the insert **104** facing the upper wall **112** of the mesh **102**, the outer surface **126** of the insert **104** facing the lower wall **114** of the mesh **102**, the distal end **128** of the insert **104** positioned near the distal portion **108** of the mesh **102**, and the proximal end **130** of the insert **104** positioned near the proximal portion **110** of the mesh **102**. The proximal end **130** of the insert **104** can be positioned distal to the hub **122**, or can extend at least partially into the lumen of the hub **122** and/or can be attached to the hub **122**. In some embodiments, the insert **104** is encapsulated within the mesh **102** but is not attached to the mesh **102**, e.g., the distal end **128** and proximal end **130** of the insert **104** remain free. Alternatively, the insert **104** can be affixed to the mesh **102** at one or both ends via adhesives, welding, bonding, fasteners, etc.

In the illustrated embodiment, because the mesh **102** is bowl-shaped with a wider distal portion **108** and a narrower proximal portion **110**, the insert **104** also has a flared (e.g., conical or bowl-like) shape with a wider distal end **128** and a narrower proximal end **130**. The flared shape can be advantageous for conforming to the geometry of the mesh **102**, as well as for promoting directional release of active agents from the insert **104**, as described further below. In other embodiments, however, depending on the shape of the mesh **102**, the distal end **128** can be narrower than the proximal end **130**, or can have the same width as the proximal end **130**. Additionally, the insert **104** can have a different shape than the flared shape shown in FIGS. **1A-1D**, including, but not limited to, tubes, cylinders, hemispheres, polyhedrons, spheroids, ovoids, plates, toruses, etc.

The insert **104** can include at least one opening exposing the inner surface **124** for interactions with materials within the aneurysm sac. For example, the distal end **128** of the insert **104** can define a distal aperture **132** to provide a passageway for blood to contact the inner surface **124** and/or for active agents released from the inner surface **124** to elute into the aneurysm cavity, as described further below. The distal aperture **132** can also receive and surround the concave portion of the upper wall **112** of the mesh **102**, such that the cavity **120** of the mesh **102** extends at least partially into the insert **104**. The proximal end **130** can also define a proximal aperture **134** (FIG. **1C**) that forms a lumen enabling material (e.g., blood, a liquid embolic) to pass through, as discussed further below. In other embodiments, however, the distal end **128** and/or proximal end **130** of the insert **104** can be closed. For example, FIG. **1D** illustrates another embodiment of the insert **104** in which the proximal end **130** does not include the proximal aperture **134**. This configuration can be advantageous for controlling the directional release of active agents eluting from the inner surface **124** (e.g., to inhibit elution out of the aneurysm and into the parent vessel), as described further detail below.

In some embodiments, the insert **104** includes at least one thrombogenic feature configured to promote thrombosis within the aneurysm sac. For example, the thrombogenic feature can be a physical structure of the insert **104** that promotes blood coagulation on and/or near the insert **104**. In some embodiments, the physical structure increases the surface area of the insert **104**, which can enhance thrombosis by providing more exposed surfaces for interaction with blood components (e.g., via physical adsorption and/or chemical bonding). Alternatively or in combination, the physical structure can disrupt blood flow through and/or near the insert **104**, which can promote hemostasis and coagulation. Examples of thrombogenic physical structures include, but are not limited to, pores, texturing, protrusions (e.g., bumps, ridges), indentations (e.g., recesses, grooves), and/or woven or braided filaments, and are described further below with respect to FIGS. **2A-2D**.

As another example, the insert **104** can be made partially or entirely of a material having thrombogenic properties, such that the material itself is the thrombogenic feature. The material can be a polymeric material, a metallic material, or a suitable combination thereof. Examples of thrombogenic materials include, but are not limited to, expanded polytetrafluoroethylene (ePTFE), etched polytetrafluoroethylene, polyurethane, polycarbonate, polyester (e.g., Dacron), polylactic acid, polyglycolic acid, polyolefin, Parylene, silicone, Cobalt Chromium (CoCr), Platinum, Tungsten, stainless steel, Nitinol, fibrin, fibronectin, gelatin, collagen, alginate, chitin, chitosan, cellulose, methylcellulose, silk, poly-N-acetylglucosamine, or combinations (e.g., mixtures, copolymers, alloys) thereof. Optionally, the thrombogenic material can include a substrate (e.g., a polymer or a metal) that has been modified with an immobilized thrombogenic agent. For example, thrombogenic agents can be applied to the surface of the substrate via coating (e.g., spray coating, dip coating, spin coating), chemical crosslinking, adsorption, deposition (e.g., chemical vapor deposition, plasma vapor deposition), or other suitable techniques known to those of skill in the art. Examples of thrombogenic agents include any of the thrombogenic materials listed above, as well as coagulation factors (e.g., prothrombin, thrombin, fibrinogen, fibrin, Factor V, Factor Va, Factor VII, Factor VIIa, Factor VIII, Factor VIIIa, Factor IX, Factor IXa, Factor X, Factor Xa, Factor XI, Factor Xia, Factor XII, Factor XIIa, Factor XIII, Factor XIIIa, tissue factor, von Willebrand factor, platelet activating factor, plasminogen activator inhibitor 1 (PAI-1)), anti-fibrinolytic agents (e.g., tranexamic acid, aminocaproic acid, aprotinin, pepstatin, leupeptin, antipain, chymostatin, gabexate), or combinations thereof. The substrate itself can also have thrombogenic properties, or can be made of a non-thrombogenic material.

In a further example, the thrombogenic feature can be a releasable thrombogenic agent that is incorporated into the insert **104**. The releasable thrombogenic agent can be or include any of the thrombogenic agents previously described herein (e.g., coagulation factors, anti-fibrinolytic agents, etc.). When the insert **104** is implanted within the aneurysm, the thrombogenic agent can elute out of the insert **104** and into the surrounding physiological environment to promote coagulation within the aneurysm sac. The thrombogenic agent can be incorporated into the insert **104** using various techniques. For example, the thrombogenic agent can be encapsulated or otherwise contained in a controlled release coating applied to the inner surface **124** and/or outer surface **126** of the insert **104**. Alternatively or in combination, the thrombogenic agent can be incorporated into the bulk material of the insert **104**, with the bulk material being configured for controlled release of the thrombogenic agent.

Optionally, the insert **104** can be configured to release other types of active agents into the surrounding environment, in addition or alternatively to the thrombogenic agent. For example, the active agent can promote healing and/or endothelialization on or near the device **100**. The active agent can be encapsulated or otherwise contained in a controlled release coating applied to the inner surface **124** and/or outer surface **126** of the insert **104**. Alternatively or in combination, the active agent can be incorporated into the bulk material of the insert **104**, with the bulk material being configured for controlled release of the active agent. Additional examples and details of techniques for achieving controlled release of thrombogenic agents and other active agents from the insert **104** are described below with respect to FIGS. **3A-3C**.

In some embodiments, one or more sections of the device **100** include a coating **136** configured to inhibit thrombogenesis. For example, when the device **100** is deployed within the aneurysm, certain components may protrude into the parent vessel and/or be exposed to blood flow from the parent vessel, such as the proximal portion **110** of the mesh **102** and/or the hub **122**. Accordingly, these components can include an anti-thrombogenic coating **136** (FIG. **1A**) configured to improve hemocompatibility and reduce thrombogenic surface activity. For example, the coating **136** can include a phosphorylcholine compound, such as Shield Technology™ (Medtronic). Additional examples of hemocompatible materials that can be used for the coating **136** include, but are not limited to, heparin, albumin, poly (ethylene oxide), immobilized and/or releasable anti-thrombogenic agents (e.g., direct thrombin inhibitors, anti-platelet agents, urokinase, tissue plasminogen activator (t-PA)), or combinations thereof. In some embodiments, the anti-thrombogenic coating **136** is localized only to those portions of the device **100** that are exposed to and/or near the parent vessel, and the remaining portions of the device **100** do not include any anti-thrombogenic coating.

Optionally, the device **100** can include at least one coating that performs other functions, such as promoting endothelialization and/or healing (e.g., via incorporation of immobilized and/or releasable vascular endothelial growth factor (VEGF), RGD peptide, etc.). The coating can alternatively or additionally increase lubricity to reduce the force needed to push the device **100** through the delivery catheter and/or withdraw the device **100** into the delivery catheter. The coating may be the same as the coating **136** (e.g., the coating **136** exhibits other properties in addition to anti-thrombogenic properties), or can be a different coating (e.g., another coating layered on top of or below the coating **136**, or on a different portion of the device **100**).

In some embodiments, the device **100** includes one or more radiopaque components to facilitate visualization. For example, some or all of the filaments of the mesh **102** can be drawn-filled tubes ("DFT") having a radiopaque core (e.g., Platinum) surrounded by a shape memory alloy and/or superelastic alloy (e.g., Nitinol, CoCr, etc.). As another example, the insert **104** can include one or more radiopaque markers **138** (FIG. **1**C), such as radiopaque beads, discs, wires, etc. The radiopaque markers **138** can be located on the outer surface **126** and/or the inner surface **124** of the insert **104**. Optionally, the radiopaque markers **138** can protrude outward from the surface(s) of the insert **104** to promote thrombogenesis via disruption of blood flow, as described elsewhere herein. Although the radiopaque markers **138** are depicted as being distributed around the periphery of the distal end **128** of the insert **104**, the radiopaque markers **138** can alternatively or additionally be located at other portions of the insert **104**, such as at the proximal end **130**, or at an intermediate region spaced apart from the distal end **128** and proximal end **130**. Optionally, radiopaque markers can alternatively or additionally be located at or near the hub **122**, on the mesh **102**, and/or any other suitable portion of the device **100**.

The device **100** can be manufactured using various techniques. For example, the mesh **102** can be formed from a braid or stent that is wrapped around a mold and heat set into the desired shape. If any coatings are used (e.g., coating **136**), the coating(s) can be applied to the specified portion(s) of the mesh **102** via dipping, spraying, deposition, etc., either before or after the mesh **102** has been shaped. The insert **104** can be fabricated separately by forming a substrate (e.g., a film, sheet, strip, mesh, membrane) into the desired shape. Optionally, in embodiments where the insert **104** is made from a shape memory material, the insert **104** can be heat set into the desired shape. In some embodiments, the substrate already has thrombogenic properties, e.g., the substrate already includes thrombogenic physical features and/or is made of a thrombogenic material. Alternatively or in combination, the substrate can be modified to include thrombogenic properties, such as by adding thrombogenic physical features (e.g., via etching, cutting, abrading), and/or by adding immobilized and/or releasable thrombogenic agents (e.g., by coating, deposition, crosslinking). The modifications to the substrate can be made before and/or after forming the substrate into the desired shape for the insert **104**.

In some embodiments, the device **100** is assembled by placing the insert **104** into the mesh **102**, after the mesh **102** has been heat set into the desired shape. Alternatively, the insert **104** can be placed into the mesh **102** before the mesh **102** has been heat set. In such embodiments, the heat setting process can be performed at a sufficiently low temperature to avoid degrading any active agents that are incorporated into the insert **104**. Once the insert **104** is positioned within the mesh **102**, the proximal portion **110** of the mesh **102** can be crimped together via the hub **122** to enclose and secure the insert **104**.

FIGS. **2**A-**2**D illustrate representative examples of thrombogenic physical features that may be incorporated into the insert **104** of FIGS. **1**A-**1**D, in accordance with embodiments of the present technology. Any of the embodiments described in connection with FIGS. **2**A-**2**D can be combined with each other and/or incorporated into any of the devices described herein.

FIG. **2**A is a perspective view of an insert **104** including a plurality of pores **202** configured to promote thrombosis, in accordance with embodiments of the present technology. The pores **202** can be formed in a substrate **204** (e.g., a continuous material such as a film, strip, sheet, membrane, etc.) via laser-cutting, etching, punching, or other suitable techniques. Alternatively or in combination, the substrate **204** itself can be made of a porous material (e.g., a polymeric material such as ePTFE), such that no additional process steps are needed to form the pores **202**. The pores **202** can have any suitable geometry (e.g., size, shape). For example, the average diameter of the pores **202** can be within a range from 0.1 μm to 1000 μm, such as from 10 μm to 300 μm. In some embodiments, the average diameter can be less than or equal to 1000 μm, 900 μm, 800 μm, 700 μm, 600 μm, 500 μm, 400 μm, 300 μm, 200 μm, 100 μm, 90 μm, 80 μm, 70 μm, 60 μm, 50 μm, 40 μm, 30 μm, 20 μm, 10 μm, or 5 μm. The pores **202** can be circular, oval, triangular, square, rectangular, diamond-shaped, polygonal, or any other regular or irregular shape, and can extend between the inner surface **124** and outer surface **126** of the insert **104**. Although the pores **202** are depicted in FIG. **2**A as having a uniform geometry (e.g., size, shape), in other embodiments, some or all of the pores **202** can have different geometries. Additionally, although the pores **202** are shown as being evenly distributed across the inner surface **124** and outer surface **126** of the insert **104**, the pores **202** can alternatively be localized to certain portions of the insert **104**, such as at or near the distal end **128** of the insert **104**, at or near the proximal end **130** of the insert **104**, at an intermediate region of the insert **104** that is spaced apart from the distal end **128** and/or proximal end **130**, etc. Moreover, although the pores **202** are depicted as being arranged in a series of circumferential rows, in other embodiments, the pores **202** can be arranged in a different pattern or can be randomly distributed.

FIG. **2**B is a perspective view of an insert **104** including a plurality of grooves **206** configured to promote thrombosis, in accordance with embodiments of the present technology. The grooves **206** can be formed in a substrate **204** (e.g., a continuous material such as a film, strip, sheet, etc.) via etching, cutting, or other suitable techniques. The grooves **206** can have any suitable geometry. For example, although the grooves **206** are illustrated as having an undulating and/or serpentine shape, in other embodiments, some or all of the grooves **206** can have a different shape, such as a zigzag shape, a linear shape, a curvilinear shape, etc. The grooves **206** can be annular structures that extend around the entire circumference of the insert **104**, or can be extend only partially around the circumference of the insert **104**. Additionally, although FIG. **2**B shows grooves **206** on both the inner surface **124** and outer surface **126** of the insert **104**, the grooves **206** can alternatively be localized to certain portions of the insert **104**, such as on the inner surface **124** only, on the outer surface **126** only, at or near the distal end **128** of the insert **104**, at or near the proximal end **130** of the insert **104**, at an intermediate region of the insert **104** that is spaced apart from the distal end **128** and/or proximal end **130**, etc. Moreover, although FIG. **2**B shows a series of horizontal grooves **206** in other embodiments, the grooves **206** can be arranged in a different pattern (e.g., vertical, angled) or can be randomly distributed.

FIG. **2**C is a perspective view of an insert **104** including texturing **208** configured to promote thrombosis, in accordance with embodiments of the present technology. The texturing **208** can be formed in a substrate **204** (e.g., a continuous material such as a film, strip, sheet, membrane, etc.) via etching, abrading, or other suitable techniques that increase the surface roughness of the substrate **204**. Alternatively, the substrate **204** itself can be made of a relatively rough material, such that no additional process steps are needed to form the texturing **208**. Although FIG. **2**C shows uniform texturing **208** on both the inner surface **124** and outer surface **126** of the insert **104**, the texturing **208** can alternatively be localized to certain portions of the insert **104**, such as on the inner surface **124** only, on the outer surface **126** only, at or near the distal end **128** of the insert **104**, at or near the proximal end **130** of the insert **104**, at an intermediate region of the insert **104** that is spaced apart from the distal end **128** and/or proximal end **130**, etc.

FIG. **2**D is a perspective view of an insert **104** formed from a mesh including a plurality of filaments **210**, in accordance with embodiments of the present technology. In some embodiments, the insert **104** is a braid and the filaments **210** are woven or braided together. Alternatively, the insert **104** can be a stent and the filaments **210** can be laser-cut struts. As shown in FIG. **2**D, the insert **104** includes a plurality of gaps **212** between the individual filaments **210**. The presence of filaments **210** interspersed with gaps **212** can disrupt blood flow through and around the insert **104**, thus promoting thrombogenesis. The filaments **210** can have any suitable size, such as a diameter within a range from 0.0004 inches to 0.0020 inches, or from 0.0009 inches to 0.0012 inches. For example, some or all of the filaments **210** can have a diameter of no more than 0.0004 inches, 0.0005 inches, 0.0006 inches, 0.0007 inches, 0.0008 inches, 0.0009 inches, 0.001 inches, 0.0011 inches, 0.0012 inches, 0.0013 inches, 0.0014 inches, 0.0015 inches, 0.0016 inches, 0.0017 inches, 0.0018 inches, 0.0019 inches, or 0.0020 inches. Some or all of the filaments **210** can have the same diameter, or some or all of the filaments **210** can have different cross-sectional diameters.

FIGS. **3**A-**3**C illustrate representative examples of controlled release coatings that may be incorporated into the insert **104** of the device **100** of FIGS. **1**A-**1**D, in accordance with embodiments of the present technology. Any of the embodiments described in connection with FIGS. **3**A-**3**C can be combined with each other and/or incorporated into any of the devices described herein.

FIG. **3**A is a side cross-sectional view of a device **100** including an insert **104** with a coating **302** on the inner surface **124** of the insert **104** ("inner coating **302**"), in accordance with embodiments of the present technology. The inner coating **302** is configured to release at least one active agent into the surrounding environment, such as a pharmaceutical compound, protein, peptide, antibody, nucleic acid, cell, particle, etc. For example, the active agent can be or include a thrombogenic agent, such as a thrombogenic material (e.g., in particulate form), coagulation factor, anti-fibrinolytic agent, or a combination thereof. As shown in FIG. **3**A, because the flared shape of the insert **104** causes the inner surface **124** of the insert **104** to be oriented generally toward the dome D of the aneurysm A, the active agent contained in the inner coating **302** can elute primarily in a distal direction toward the aneurysm dome D and away from the aneurysm neck N. In embodiments where the active agent is a thrombogenic agent, this configuration can be advantageous for ensuring that thrombogenesis occurs primarily or entirely within the cavity of the aneurysm, rather than near or within the parent vessel V.

Although FIG. **3**A illustrates a single inner coating **302**, in other embodiments, the insert **104** can include multiple coatings **302** on the inner surface **124**, such as two, three, four, five, or more layers of inner coatings **302**. Some or all of the inner coatings **302** can include the same active agent, or some or all of the inner coatings **302** can include different active agents. Optionally, some of the inner coatings **302** may not include any active agent, and can instead serve other functions (e.g., enhancing adhesion of other coating layers, providing protection against damage, etc.). The outer surface **126** of the insert **104** can remain uncoated, or can also include one or more coatings, as discussed further below.

FIG. **3**B is a side cross-sectional view of a device **100** including an insert **104** with a coating **304** on the outer surface **126** of the insert **104** ("outer coating **304**"), in accordance with embodiments of the present technology. The outer coating **304** is configured to release at least one active agent into the surrounding environment, such as a pharmaceutical compound, protein, peptide, antibody, nucleic acid, cell, particle, etc. For example, the active agent can be or include an agent that inhibits thrombogenesis (e.g., heparin, albumin, direct thrombin inhibitors, anti-platelet agents, urokinase, t-PA) and/or promotes endothelialization and/or healing (e.g., EGF, RGD peptide). As shown in FIG. **3**B, because the flared shape of the insert **104** causes the outer surface **126** of the insert **104** to be oriented generally toward the neck N of the aneurysm A, the active agent contained in the outer coating **302** can elute primarily in a proximal direction toward the aneurysm neck N and away from the aneurysm dome D. In embodiments where the active agent is an anti-thrombogenic, endothelialization, and/or healing agent, this configuration can be advantageous for preventing thrombosis in the parent vessel V and/or promoting endothelialization and/or healing across the neck N of the aneurysm.

Although FIG. **3**B illustrates a single outer coating **304**, in other embodiments, the insert **104** can include multiple coatings **304** on the outer surface **126**, such as two, three, four, five, or more layers of outer coatings **304**. Some or all of the outer coatings **304** can include the same active agent, or some or all of the outer coatings **304** can include different active agents. Optionally, some of the outer coatings **304** may not include any active agent, and can instead serve other functions (e.g., enhancing adhesion of other coating layers, providing protection against damage, etc.). The inner surface **124** of the insert **104** can remain uncoated, or can also include one or more coatings, as described above.

FIG. **3**C is a side cross-sectional view of a device **100** including an insert **104** with both an inner coating **302** and an outer coating **304**. Each of the coatings **302**, **304** is configured to release at least one respective active agent into the surrounding environment. In some embodiments, the inner coating **302** is configured to release the same active agent(s) as the outer coating **304**, e.g., both coatings **302**, **304** release a thrombogenic agent. Alternatively, the inner coating **302** can release a different active agent or agents than the outer coating **304**, e.g., the inner coating **302** releases a thrombogenic agent while the outer coating **304** releases an anti-thrombogenic, endothelialization, and/or healing agent. Accordingly, the insert **104** can promote occlusion of the aneurysm sac, while concurrently facilitating healing of the aneurysm neck to inhibit recanalization. Although FIG. **3**C illustrates a single inner coating **302** and a single outer coating **304**, in other embodiments, the insert **104** can include multiple coatings on the inner surface **124** and/or outer surface **126**, as previously described.

The controlled release coatings described herein can be fabricated in various ways. For example, any of the coatings herein can include at least one active agent mixed with at least one polymeric material configured to mediate release of the active agent. In some embodiments, the active agent is released as the polymeric material degrades within the physiological environment of the aneurysm. Alternatively or in combination, the polymeric material can include pores, interstices, spaces, etc., that enable the active agent to elute out over time, with or without degradation of the polymeric material itself. Examples of suitable polymeric materials include, but are not limited to, BioLinx® (Medtronic), poly(vinyl alcohol), poly(ethylene-vinyl acetate), polyurethane, polycaprolactone, polyglycolic acid, polylactic acid, poly(lactide-co-glycolide), poly(ethylene oxide), poly(vinyl pyrrolidone), silicone, a silicone-urethane copoplymer, an acrylic polymer, an acrylic-acrylonitrile copolymer, a latex polymer, or combinations thereof. Optionally, the coating can include the active agent by Itself without any polymeric material. The coatings disclosed herein can be applied to the insert **104** using any suitable technique, such as dip coating, spraying coating, spin coating, adsorption, deposition, covalent bonding via a degradable crosslinker, etc.

Alternatively or in combination, the insert **104** itself can be made partially or entirely of a bulk material configured to provide controlled release of at least one active agent. The bulk material can include the active agent(s) mixed with a polymeric material, such as any of the materials described above in connection with controlled release coatings. In some embodiments, the bulk material is a degradable material, such that the insert **104** breaks down over time after implantation in the aneurysm. In other embodiments, the bulk material can be a nondegradable material, such that the insert **104** remains permanently within the aneurysm.

The insert **104** can be configured to provide controlled release of any suitable number of active agents, such as one, two, three, four, five, or more different active agents. The active agent(s) can be continuously released within the aneurysm over any suitable time period, such as a time period of at least 1 hour, 2 hours, 4 hours, 12 hours, 1 day, 2 days, 3 days, 4 days, 5 days, 6 days, 1 week, 2 weeks, 3 weeks, 4 weeks, 1 month, 2 months, 3 months, 6 months, or a year. The timing of the controlled release can be tuned based on factors such as the type of polymeric material used, the type of active agent(s) used, the amount of the active agent(s), the geometry of the insert **104**, the thickness of the coating(s) and/or bulk material, or combinations thereof.

FIGS. **4**A and **4**B illustrate another occlusive device **400** ("device **400**") configured in accordance with embodiments of the present technology. The device **400** can be generally similar to the device **100** of FIGS. **1**A-**1**D. Accordingly, like numbers (e.g., mesh **102** versus mesh **402**) are used to identify similar or identical structures, and discussion of the device **400** of FIGS. **4**A and **4**B will be limited to those features that differ from the device **100** of FIGS. **1**A-**1**D. Additionally, any of the features of the devices **100** and **400** described herein can be combined with each other.

Referring to FIG. **4**A, which illustrates a perspective view of the device **400**, the device **400** includes a mesh **402** having an annular, bowl-shaped body **406** configured to be positioned within the aneurysm sac, a distal portion **408** configured to be oriented toward the aneurysm dome, and a proximal portion **410** configured to cover the aneurysm neck. In the illustrated embodiment, the distal portion **408** of the mesh **402** has an undulating shape defining a plurality of petals **440** ("mesh petals **440**"). Stated differently, the distal end of the upper wall **412** of the mesh **402**, the distal end of the lower wall **414** of the mesh **402**, and the fold region **418** connecting the upper wall **412** and lower wall **414** can collectively define the structure of each mesh petal **440**. The distal end of the upper wall **412** and the distal end of the lower wall **414** can be separated by a gap, such that each mesh petal **440** is a hollow structure defining a distal end sub-cavity **441** that is connected to the internal cavity **416** of the mesh **402**. The mesh petals **440** can be peaks, flanges, protrusions, etc., that surround the cavity **420** of the mesh **402** to facilitate retention of an embolization element within the cavity **420**. Although FIG. **4**A illustrates the mesh **402** as having seven mesh petals **440**, the mesh **402** can alternatively have a different number of mesh petals **440**, such as two, three, four, five, six, eight, nine, ten, or more mesh petals **440**.

As best seen in FIG. **4**B, which shows a perspective view of the insert **404** of the device **400**, the distal end **428** of the insert **404** can also include a respective plurality of petals **442** ("insert petals **442**") surrounding the distal aperture **432** of the insert **404**. In the illustrated embodiment, the insert petals **442** are configured similarly to the mesh petals **440** (e.g., with respect to shape, size, number, and/or location). Accordingly, when the insert **404** is positioned within the mesh **402** (FIG. **4**A), each insert petal **442** is aligned with and/or received at least partially within the distal end sub-cavity **441** of a corresponding mesh petal **440**. In other embodiments, however, the insert **404** can include a different number of petals than the mesh **402**. For example, although the insert **404** is depicted as having seven insert petals **442**, the insert **404** can alternatively have two, three, four, five, six, eight, nine, ten, or more petals **442**. Additionally, some or all of the insert petals **442** can be circumferentially offset from the mesh petals **440**. Optionally, the insert **404** can have a different shape than the mesh **402**. For example, the mesh **402** can include petals **440** while the insert **404** does not include any petals (e.g., similar to the insert **104** of FIGS. **1**A-**1**D), or the insert **404** can include petals **442** while the mesh **402** does not include any petals (e.g., similar to the mesh **102** of FIGS. **1**A and **1**B).

FIGS. **5**A-**5**E illustrate a method of treating an aneurysm with an occlusive device, in accordance with embodiments of the present technology. Although the illustrated embodiment is shown and described in terms of the device **100** of FIGS. **1**A-**1**D, the method can be applied to any embodiment of the occlusive devices described herein (e.g., the device **400** of FIGS. **4**A and **4**B).

Referring first to FIG. **5**A, the device **100** can be loaded within a first elongate shaft **502** (e.g., a delivery catheter such as a microcatheter) in a low-profile configuration. When in the low-profile configuration, the mesh **102** and insert **104** of the device **100** can be compressed, flattened, or otherwise compacted in a generally linear configuration to conform to the interior lumen of the first elongate shaft **502**.

The device **100** can then be intravascularly delivered to a location within a blood vessel V adjacent a target aneurysm A via the first elongate shaft **502**. As shown in FIG. **2**A, a distal tip **504** of the first elongate shaft **502** can be advanced through the neck N of the aneurysm A and into an interior cavity of the aneurysm A. The device **100** can then be deployed by pushing the device **100** distally through the opening in the distal tip **504** of the first elongate shaft **502** and into the aneurysm cavity, e.g., using a pusher member (not shown) coupled to the hub **122** of the device **100**.

Referring next to FIG. **5**B, as the device **100** exits the first elongate shaft **502**, the mesh **102** can self-expand from the low-profile configuration into the expanded configuration. The self-expansion forces from the mesh **102** can push the insert **104** into its expanded configuration, or the insert **104** can also self-expand when released from the constraints of the first elongate shaft **502**. As shown in FIG. **5**B, in the expanded configuration, the upper wall **112** and cavity **120** of the mesh **102** face the dome D of the aneurysm A, while the lower wall **114** of the mesh **102** contacts the interior surface of the aneurysm A and bridges the neck N. Once the device **100** is deployed, the hub **122** can be detached from the pusher member, and the pusher member and first elongate shaft **502** can then be withdrawn from the aneurysm A.

Referring next to FIG. **5**C, blood entering the aneurysm A from the parent vessel V interacts with the thrombogenic feature(s) of the insert **104**, resulting in formation of thrombus **506** on and/or near the insert **104**. As previously described, the thrombogenic feature(s) can include any of the following: (a) a physical structure that increases the surface area of the insert **104** and/or disrupts blood flow near the insert **104**, such as any of the embodiments of FIGS. **2**A-**2**D; (b) a thrombogenic material used to form all or a portion of the insert **104**, such as ePTFE or CoCr; and/or (c) a releasable thrombogenic agent incorporated into the insert **104**, e.g., in accordance with the embodiments of FIGS. **3**A-**3**C. Optionally, the device **100** can include an anti-thrombogenic coating **136** on components that protrude into the parent vessel V, such as the proximal portion of the mesh **102** and/or the hub **122**.

Referring next to FIG. **5**D, the thrombogenic feature(s) of the insert **104** can promote hemostasis and/or coagulation within the aneurysm A until most or all of the cavity is filled with thrombus **506**. The portions of the device **100** including the anti-thrombogenic coating **136** can remain partially or completely free of thrombus **506**, thus providing an exposed substrate for growth of an endothelial layer.

Referring next to FIG. **5**E, the device **100** can optionally be used in combination with an embolization element, such as a liquid embolic **508**. The liquid embolic **508** can be introduced into the aneurysm A via a second elongate shaft **510** coupled to the hub **122**. The second elongate shaft **510** can be received within the first elongate shaft **502** (not shown in FIG. **5**E) and advanced to the aneurysm A together with the device **100**. After the device **100** is deployed, the liquid embolic **508** is pumped into the aneurysm cavity through the second elongate shaft **510**, and passes through the hub **122**, the proximal aperture **134** in the insert **104**, and the mesh **102**, and into the interior of the aneurysm A. Subsequently, the second elongate shaft **510** can be decoupled from the hub **122** and withdrawn, leaving the device **100** and liquid embolic **508** in place. The thrombogenic feature(s) of the insert **104** can function in concert with the liquid embolic **508** to occlude the interior of the aneurysm A.

Alternatively, the second elongate shaft **510** can be a separate component that is not coupled to the hub **122**, but is instead introduced into the aneurysm A before the device **100** is deployed. When the device **100** is subsequently deployed, the mesh **102** can expand to push the second elongate shaft **510** outwardly toward the side of the aneurysm A to hold the second elongate shaft **510** in place between the mesh **102** and the inner surface of the aneurysm A. The liquid embolic **508** (or other embolization element, such as a coil) can then be introduced into the aneurysm A via the second elongate shaft **510**.

CONCLUSION

Although many of the embodiments are described above with respect to systems, devices, and methods for treating a cerebral aneurysm, the technology is applicable to other applications and/or other approaches. For example, the occlusive devices, systems, and methods of the present technology can be used to treat any vascular defect and/or fill or partially fill any body cavity or lumen or walls thereof, such as for parent vessel take down, endovascular aneurysms outside of the brain, arterial-venous malformations, embolization, atrial and ventricular septal defects, patent ductus arteriosus, and patent foramen ovale. Additionally, several other embodiments of the technology can have different states, components, or procedures than those described herein. It will be appreciated that specific elements, substructures, advantages, uses, and/or other features of the embodiments described can be suitably interchanged, substituted or otherwise configured with one another in accordance with additional embodiments of the present technology. A person of ordinary skill in the art, therefore, will accordingly understand that the technology can have other embodiments with additional elements, or the technology can have other embodiments without several of the features shown and described above with reference to FIGS. **1**A-**5**E.

The descriptions of embodiments of the technology are not intended to be exhaustive or to limit the technology to the precise form disclosed above. Where the context permits, singular or plural terms may also include the plural or singular term, respectively. Although specific embodiments of, and examples for, the technology are described above for illustrative purposes, various equivalent modifications are possible within the scope of the technology, as those skilled in the relevant art will recognize. For example, while steps are presented in a given order, alternative embodiments may perform steps in a different order. The various embodiments described herein may also be combined to provide further embodiments.

As used herein, the terms "generally," "substantially," "about," and similar terms are used as terms of approximation and not as terms of degree, and are intended to account for the inherent variations in measured or calculated values that would be recognized by those of ordinary skill in the art.

Moreover, unless the word "or" is expressly limited to mean only a single item exclusive from the other items in reference to a list of two or more items, then the use of "or" in such a list is to be interpreted as including (a) any single item in the list, (b) all of the items in the list, or (c) any combination of the items in the list. As used herein, the phrase "and/or" as in "A and/or B" refers to A alone, B alone, and A and B. Additionally, the term "comprising" is used throughout to mean including at least the recited feature(s) such that any greater number of the same feature and/or additional types of other features are not precluded.

It will also be appreciated that specific embodiments have been described herein for purposes of illustration, but that various modifications may be made without deviating from the technology. Further, while advantages associated with certain embodiments of the technology have been described in the context of those embodiments, other embodiments may also exhibit such advantages, and not all embodiments need necessarily exhibit such advantages to fall within the scope of the technology. Accordingly, the disclosure and associated technology can encompass other embodiments not expressly shown or described herein.

I claim:

1. An occlusive device for treating an aneurysm, the occlusive device comprising:

an expandable mesh configured to span a neck of the aneurysm, the expandable mesh comprising an upper wall and a lower wall, wherein an internal cavity is defined between the upper wall and the lower wall, and wherein the expandable mesh comprises an anti-thrombogenic coating localized to a portion of the lower wall of the expandable mesh; and

an insert positioned within the internal cavity of the expandable mesh between the upper wall and the lower wall, the insert comprising a film having an inner surface and an outer surface, wherein the inner surface of the film faces the upper wall of the expandable mesh, and the outer surface of the film faces the lower wall of the expandable mesh and wherein the insert comprises an open upper end defining an aperture,

wherein the insert is configured to promote thrombosis within the aneurysm and the anti-thrombogenic coating is configured to discourage thrombosis across the neck of the aneurysm when the occlusive device is deployed within the aneurysm, and

wherein when the occlusive device is deployed within the aneurysm, the expandable mesh forms a bowl structure comprising a distal portion, a proximal portion, and the internal cavity extending from the distal portion toward the proximal portion.

2. The occlusive device of claim **1**, wherein, when the occlusive device is deployed within the aneurysm, the insert comprises a flared shape comprising a proximal end, and a distal end wider than the proximal end.

3. The occlusive device of claim **1**, wherein the insert comprises a polymeric material, a metallic material, or a combination thereof.

4. The occlusive device of claim **1**, wherein the insert defines a plurality of pores extending between the inner surface and the outer surface of the film, and wherein the plurality of pores are configured to enhance thrombogenicity of the insert.

5. The occlusive device of claim **1**, wherein the insert comprises texturing on one or more of the inner surface or the outer surface of the film, and wherein the texturing is configured to enhance thrombogenicity of the insert.

6. The occlusive device of claim **1**, wherein the insert is configured to release at least one active agent into the aneurysm.

7. The occlusive device of claim **6**, wherein the at least one active agent is configured to promote thrombosis within the aneurysm.

8. An occlusive device for treating an aneurysm, the occlusive device comprising:

a mesh configured to self-expand into a bowl structure, the mesh including a distal layer and a proximal layer, wherein the distal layer comprises a concave surface defining a cavity of the bowl structure; and

an insert positioned between the distal layer and the proximal layer, wherein the insert comprises a membrane having an outer surface and an inner surface, wherein the insert comprises an open distal end defining an aperture, and wherein the insert comprises at least one thrombogenic feature.

9. The occlusive device of claim **8**, further comprising a coating on a portion of the proximal layer of the mesh, wherein the coating is configured to reduce thrombogenesis on the portion of the proximal layer.

10. The occlusive device of claim **8**, wherein the outer surface of the membrane is oriented toward the proximal layer of the mesh, and the inner surface of the membrane is oriented toward the distal layer of the mesh.

11. The occlusive device of claim **8**, wherein the insert comprises a flared shape comprising a proximal end that is narrower than the distal end.

12. The occlusive device of claim **11**, wherein the narrower proximal end of the insert defines a proximal aperture configured to allow material to pass therethrough.

13. The occlusive device of claim **8**, wherein the at least one thrombogenic feature comprises one or more of a thrombogenic material, a plurality of pores, texturing on one or more of the inner surface or the outer surface of the membrane, a plurality of filaments, or a releasable agent.

14. An occlusive device for treating an aneurysm, the occlusive device comprising:

an expandable mesh configured to span a neck of the aneurysm, the expandable mesh comprising a distal wall and a proximal wall, wherein when the occlusive device is deployed within the aneurysm, the expandable mesh forms a bowl structure comprising the distal wall, the proximal wall, and an internal cavity between the distal wall and proximal wall; and

an insert positioned in the internal cavity, wherein the insert is formed from a single material layer having an inner surface and an outer surface, wherein the insert comprises an open distal end defining an aperture, and wherein the insert includes at least one thrombogenic feature.

15. The occlusive device of claim **14**, wherein the insert comprises a mesh.

16. The occlusive device of claim **14**, wherein the single material layer comprises a thrombogenic material, and the at least one thrombogenic feature comprises the thrombogenic material.

17. The occlusive device of claim **14**, wherein the at least one thrombogenic feature comprises a physical structure formed in the single material layer of the insert, wherein the physical structure comprises pores, texturing, protrusions, indentations, or a combination thereof.

18. The occlusive device of claim **14**, wherein the at least one thrombogenic feature comprises a releasable thrombogenic agent.

19. The occlusive device of claim **18**, wherein the releasable thrombogenic agent is (a) incorporated into a coating on one or more of the inner surface or the outer surface of the single material layer, (b) incorporated into the single material layer of the insert, or (c) both (a) and (b).

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